Macroeconomic Sentiments and Job Search Behavior

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Abstract

Employed workers who expect the labor market to worsen perceive a greater risk of losing their own jobs and search more intensively while employed. Using survey data, I show that this relationship is driven by workers' fear of job loss rather than by their expectations about finding a new job. I interpret the pattern as precautionary on-the-job search: workers look for outside offers to protect themselves against the possibility of unemployment. To quantify this channel, I embed the behavior in a search-and-matching model that separates precautionary search from ordinary job-ladder search. The model matches the empirical search response, offer receipt, total employer-to-employer mobility, and the share of employer-to-employer moves related to job loss. The calibrated model implies that job-loss beliefs matter for how much employed workers search and for the cyclicality of that search.

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1 Introduction

Do households adjust their job search effort before labor market conditions worsen? Recessions make job search harder: jobs are scarcer, more workers are looking, and unemployment risk is higher. If households foresee an economic downturn, they may also perceive a higher probability of losing their own job and search while still employed to mitigate this risk. Using individual-level survey data from the United States, I show that employed workers who are pessimistic about the economy also expect higher risk of job loss and search more on the job. I then use these facts to discipline a search-and-matching model and quantify what they imply for search behavior and unemployment flows.

This paper makes three contributions. First, I show that pessimism about the economy and perceived job-loss risk predict an increase in search intensity among employed workers. Second, I show that the response is consistent with precautionary search. It is tied to perceived job-loss risk and job-security motives. Third, I quantify the mechanism in a Diamond–Mortensen–Pissarides model with endogenous on-the-job search (Mortensen and Pissarides, 1994).[1]

This paper makes three contributions. First, I document that employed workers who are pessimistic about the economy perceive a higher risk of losing their job and search more while employed. Second, I show that this response is precautionary: it is concentrated among workers with the most to lose from job loss and is tied to job-security motives. Third, I quantify the mechanism in a Diamond–Mortensen–Pissarides model with endogenous on-the-job search (Mortensen and Pissarides, 1994).[2]

Using the Survey of Consumer Expectations (SCE), a monthly panel that elicits probabilistic expectations alongside detailed measures of on-the-job search (Armantier et al., 2017), I document that a one standard deviation increase in macro pessimism is associated with roughly 24 additional minutes of search per week among employed workers.[3] This relatioship operates largely through personal labor-market expectations. Workers who are more pessimistic about aggregate unemployment report higher job-loss risk, while

[1]The baseline model treats employed search as a way to build outside options before job loss occurs. I then extend the model so the same offer technology can also generate ordinary job-ladder moves. This allows the model to distinguish the insurance value of outside options from the broader volume of employer-to-employer mobility.

[2]The baseline model treats employed search as a way to hold outside options before job loss occurs. I then extend the model so the same offer technology can also generate usual employer-to-employer mobility, so that survey data on the composition of job changes discipline how much employed search can affect unemployment.

[3]The paper primarily measures macro pessimism as the subjective probability that unemployment will be higher in 12 months. I use other measures of macro expectations such as credit access, interest rates, house prices, government debt and stock prices as robustness.

their expected job-finding probability conditional on job loss moves much less. The same belief-to-risk-to-search pattern also appears in the ECB Consumer Expectations Survey, providing an external validation on the mechanism.

Three pieces of evidence point to a precautionary motive. Among employed workers who search, higher perceived job-loss risk predicts a higher probability of citing job security as a search motive. The search response is also strongest among workers who report both high job-loss risk and weak re-employment prospects, the case in which an outside offer is most useful. Finally, perceived job-finding probability enters negatively: workers who expect to find work quickly if separated search *less*. This pattern is hard to reconcile with a pure job-ladder story, but it follows naturally if search partly insures against unemployment.

Reported job-loss probabilities are also predictive of later separations. Workers who assign a higher probability to losing their job are more likely to separate over the subsequent three months, and the relationship is monotone across belief quintiles. This is consistent with the idea that some workers have advance information about job loss (Hendren, 2017).[4] The same beliefs that predict search also forecast later separations. I use these facts to motivate and discipline the model.

To quantify the mechanism, I build a Diamond–Mortensen–Pissarides model in which employed workers can search before job loss occurs. The model's central feature is a wedge between the job-loss risk workers face and the risk they perceive: search decisions respond to perceived risk, while realized flows into unemployment are governed by actual separations. A worker who believes job loss is likely searches harder, is more likely to hold an outside offer when the separation arrives, and moves to another employer rather than into unemployment. Aggregate conditions however, push in opposite directions: recessions make offers harder to obtain, which discourages search, but they also raise perceived job-loss risk, which encourages it. Whether employed search rises or falls in a recession is therefore a quantitative question about beliefs. A job-ladder extension allows the same offers to generate ordinary employer-to-employer moves when the current job survives, so that data on total mobility and its composition can separate the insurance value of an offer from its job-ladder value.

The calibrated model delivers three results. First, beliefs are what make employed search countercyclical. In the baseline, workers search about 26 percent more in recessions than in expansions; if perceived job-loss risk is held fixed across aggregate states, search

[4]Advance information and precautionary behavior are distinct. I show that workers who perceive higher risk adjust search in a way that is consistent with self-insurance, especially when they are pessimistic about re-employment.

turns procyclical, because decreasing tightness makes effort less productive. Second, if every outside offer could be used at separation, this search response would be a meaningful stabilizer: holding search at its expansion level raises recession unemployment by 0.61 percentage points, one-sixth of the recessionary rise, although this estimate is an upper-bound. Third, in the SCE, workers move between employers at 3.85 percent per month, but only 9.4 percent of those moves are related to job loss, so insured transitions amount to about 0.36 percent of employment per month. Once the model matches this composition, endogenous search offsets roughly 1 percent of the recessionary rise in aggregate unemployment. The behavioral finding and the aggregate finding are therefore distinct: beliefs explain why employed search rises in recessions, while the composition of employer-to-employer moves determines how much that search can do to unemployment.

A final counterfactual isolates the decision margin directly. I raise workers' perceived job-loss risk while holding separations, tightness, and productivity fixed, so that nothing about the labor market changes except what workers believe. Search rises, offers become more common, and fewer of the workers who do separate fall into unemployment: beliefs alone move worker flows, and they do so through no channel other than precautionary search. The experiment mirrors the empirical design, which identifies the search response from variation in beliefs across workers facing the same labor market.

The paper relates to three strands of work. First, it contributes to research on subjective expectations and labor-market behavior. Workers' beliefs about labor-market prospects are informative and behaviorally relevant: subjective job-insecurity beliefs vary across workers (Manski and Straub, 2000), employed workers have advance information about job loss (Hendren, 2017), and job seekers' beliefs predict outcomes but can be biased (Spinnewijn, 2015; Conlon et al., 2018; Mueller et al., 2021). I add evidence on a different decision margin: employed workers' perceived job-loss risk predicts how much they search before any separation occurs.

Second, the paper relates to on-the-job search and job-ladder models. Standard models emphasize search for better jobs and higher wages (Burdett and Mortensen, 1998; Postel-Vinay and Robin, 2002; Menzio and Shi, 2011; Moscarini and Postel-Vinay, 2018), while recent empirical work shows that employed search is common, effective, and cyclical (Mukoyama et al., 2018; Faberman and Kudlyak, 2019; Faberman et al., 2022). Bransch (2021) studies the cyclicality of unemployed search intensity. The closest papers are Ahn and Shao (2017) and Simmons (2024), which study precautionary on-the-job search. Relative to this work, the contribution is measurement: direct subjective probabilities separate perceived separation risk from perceived re-employment prospects, stated motives identify why workers search, and the model converts these micro moments into a bound on how much employed search can matter for unemployment.

Third, the paper contributes to work on expectations and business-cycle fluctuations. Expectations can move aggregate outcomes (Beaudry and Portier, 2004; Lorenzoni, 2009; Eusepi and Preston, 2011; Barsky and Sims, 2012), including in labor-market models with matching frictions and learning (Den Haan and Kaltenbrunner, 2009; Di Pace et al., 2016). Survey expectations also matter for household decisions, and disagreement in macro beliefs is widespread (Mankiw et al., 2003; Coibion et al., 2018). In labor markets, macro perceptions can affect personal expectations and behavior (Carroll and Dunn, 1997; Roth and Wohlfart, 2019). Pilossoph and Ryngaert (2024) and Raposo (2025) show that inflation expectations affect employed search through wage-related motives. Here, macro beliefs affect employed search through perceived job-loss risk and the insurance value of outside offers.

The rest of the paper is organized as follows. Section 2 documents the empirical relationship between perceived separation risk and on-the-job search. Section 3 develops a search model with job-ladder and precautionary motives. Section 4 describes the calibration and model fit. Section 5 reports the quantitative implications. Section 6 concludes.

2 Perceived Separation Risk and On-the-Job Search

This section uses individual-level data from the Survey of Consumer Expectations (SCE) to study how employed workers' labor-market expectations relate to on-the-job search. Workers who report higher job-loss risk search more while employed. Several facts documented in this section point to a precautionary motive: reported job-loss risk predicts subsequent separations, predicts job-security search motives, and is most important when perceived re-employment prospects are weak.

The SCE analysis proceeds in four steps. Section 2.2 relates expectations to search. Section 2.3 shows that macroeconomic pessimism maps more strongly into personal job-loss expectations than into perceived job-finding prospects. Section 2.4 shows that reported job-loss probabilities forecast realized separations. Section 2.5 uses stated search motives and interaction patterns to separate precautionary search from other reasons for on-the-job search. After robustness checks, Section 2.7 asks whether the same belief-to-risk-to-search link appears in the ECB Consumer Expectations Survey.

2.1 The Survey of Consumer Expectations

I use the Survey of Consumer Expectations (SCE), a nationally representative monthly internet-based panel conducted by the Federal Reserve Bank of New York since 2013. Each month approximately 1,300 household heads complete the survey; respondents typically remain in the panel for 12 consecutive months. The SCE has two components relevant to this paper. The *core monthly survey* collects employment status and probabilistic expectations—elicited on a 0–100 scale—about aggregate and personal labor market outcomes. The *Labor Market Module* (LMM) is administered to the same respondents every four months, so each participant completes it up to three times during their panel tenure. The LMM collects detailed information on job search behavior. For further details on the SCE, see Armantier et al. (2017).

Labor Market Module. The LMM's primary search measure asks: "How many hours did you spend in the past 7 days looking or applying for jobs?" This question captures the *intensive margin* of on-the-job search. The module also records which search activities the respondent undertook (browsing postings, applying online, networking, contacting employers directly) and, for the mechanism tests in Section 2.5, the respondent's stated reasons for searching. The analysis sample consists of 4,226 employed worker-months with non-missing search hours, spanning March 2014 through December 2023.[5]

Expectation variables. The empirical analysis relies on four probabilistic expectation variables from the SCE core survey, each elicited on a 0–100 scale as a point estimate rather than a full density forecast. *Macro pessimism* is the respondent's subjective probability that "12 months from now the unemployment rate will be higher than it is now" (sample mean: 39.4%, SD: 23.9 pp). *Job-loss probability* is the percent chance the respondent assigns to involuntarily losing their current job in the next 12 months (mean: 21.3%, SD: 23.4 pp). *Voluntary separation probability* is the percent chance of voluntarily leaving the current job in the next 12 months (mean: 40.2%, SD: 30.4 pp). *Job-finding probability conditional on job loss* is the percent chance of finding a new job within three months if the current job is lost (mean: 57.1%, SD: 30.0 pp). Expected inflation is also collected in the core survey and enters the regressions as a control. Because the survey elicits point estimates, workers with the same reported probability may differ in subjective uncertainty; this heterogeneity is not observed.

[5]The full SCE Labor Market Module contains approximately 12,000 respondent-wave observations over this period; the analysis sample restricts to employed workers with non-missing search hours and expectation variables.

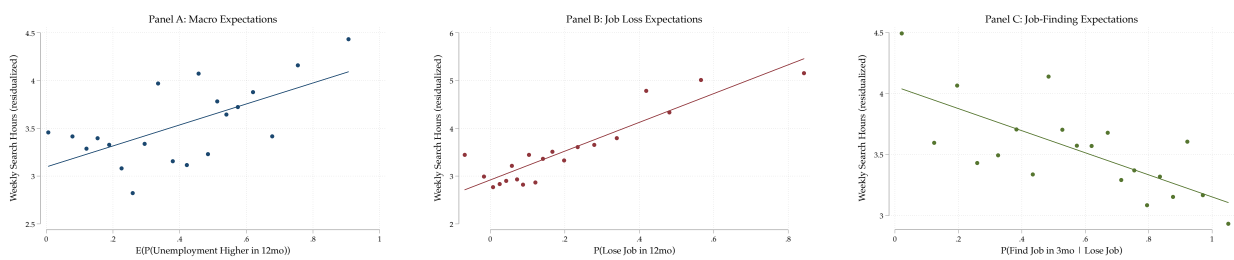
Appendix Table A1 reports summary statistics. The average employed worker in the LMM reports approximately 3.8 hours per week of search (se: 4.8 hours/week).

2.2 Perceived risk and search intensity

I begin by relating expectations to search intensity. Search is forward-looking: a worker deciding whether to look for another job cares both about aggregate conditions and about her own job. I condition on detailed controls and fixed effects, so the estimates compare workers in the same time and location environment. The baseline specification regresses weekly search hours on the worker's subjective probability that unemployment will be higher in 12 months:

$$\text{SearchHours}_{ist} = \beta\, E_{it}\big[\Delta\,\Pr(U_{t+12} \geq 0)\big] + \mathbf{X}_{it}'\gamma + \mu_s + \lambda_t + \varepsilon_{ist}, \tag{1}$$

where $E_{it}[\Delta\,\Pr(U_{t+12} \geq 0)]$ is scaled to $[0,1]$, $\mathbf{X}_{it}$ includes demographic controls, and μ_s and λ_t denote state and time fixed effects. Standard errors are clustered at the individual level. Figure 1 (Panel A) displays the conditional relationship between macro pessimism and search hours after residualizing both variables with respect to controls and fixed effects.



Macro expectations | Job-loss expectations | Job-finding expectations

Notes: Each panel plots binned means of weekly search hours against a single expectation measure after residualizing both variables on the full set of controls, state fixed effects, and time fixed effects. Expectations are scaled to $[0,1]$ and search hours are winsorized at the 5th and 95th percentiles. Sample: employed respondents in the SCE Labor Market Module.

Figure 1. Binscatter: expectations vs. search hours

Column (1) of Table 1 reports the corresponding coefficient. A one-standard-deviation increase in macro pessimism (24 percentage points) is associated with about 24 additional minutes of job search per week, roughly 10 percent of mean search. The estimate changes little when I include expectations about other macro variables, including inflation, stock prices, credit access, house prices, interest rates, and government debt. Figure 1 (Panel A) shows the corresponding conditional binscatter.

Aggregate beliefs are informative, but workers also have beliefs about their own labor-market prospects. Columns (2)–(5) of Table 1 therefore add personal expectations about job finding and separation. Two results stand out. Perceived *job-loss probability* predicts higher search intensity, while perceived *job-finding probability conditional on job loss* enters with the opposite sign. Workers who expect to find a job quickly if separated search *less*. These signs are consistent with precautionary search: an outside offer is more valuable when job-loss risk is high and unemployment would be costly. Voluntary separation expectations enter positively, consistent with a job-ladder margin;[6] the focus here is the perceived job-loss channel.

Table 1. Expectations and On-the-Job Search Intensity

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
$E\_t(\Delta\,\mathrm{Pr}(U\_t + 12 \geq 0))$	1.607***	0.953***	0.953***	0.872**	0.928***
	(0.372)	(0.356)	(0.355)	(0.370)	(0.352)
E_t(Pr(Lose Job_t + 12))		2.046***	1.836***	1.629***	2.682***
		(0.425)	(0.429)	(0.441)	(0.831)
E_t(Pr(Leave Job_t + 12))		1.567***	1.884***	1.313***	1.890***
		(0.330)	(0.333)	(0.361)	(0.334)
E_t(Pr(Find|Lose_t + 3))			-1.156***	-1.568***	-0.877**
			(0.294)	(0.298)	(0.365)
E_t(Pr(Offer_t + 3))				0.019***	
				(0.003)	
E_t(Pr(Lose_t + 3)) × E_t(Pr(Find|Lose_t + 3))					-1.479
					(1.214)
$E\_t(\pi\_t + 12)$ (dist. mean)	-0.025	-0.025	-0.028	-0.024	-0.029
	(0.021)	(0.022)	(0.022)	(0.023)	(0.022)
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	4132	4131	4130	3922	4130
R-squared	0.125	0.148	0.152	0.163	0.153

Notes: Standard errors in parentheses. Dependent variable is weekly job search hours (winsorized). Sample is employed workers. All specifications include state and time fixed effects, are weighted by survey weights, and control for age polynomial, education, race, gender, marital status, and household income. Standard errors are clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

After conditioning on personal expectations, the macro pessimism coefficient falls but remains positive. This decline suggests that macro pessimism predicts search in part

[6]The job-ladder literature emphasizes that employed workers search to climb toward better-paying jobs; see Postel-Vinay and Robin (2002) and Moscarini and Postel-Vinay (2018) for theory and evidence.

because it is related to workers' own perceived job-loss risk. The remaining coefficient may capture other aggregate risks not covered by the SCE's personal expectation questions.

To make the comparison more discrete, Table 2 repeats the same exercise using indicators for whether each of the four core expectations lies above the sample median. This specification compares workers with relatively high beliefs to those at or below the median rather than tracing marginal responses to a one-unit change in reported probabilities.

Table 2. Expectations and On-the-Job Search Intensity: Above-Median Indicators

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
Above median: $E_t(\Delta \Pr(U_{t+12} \geq 0))$	0.765^{***}	0.519^{***}	0.520^{***}	0.538^{***}	0.512^{***}
	(0.174)	(0.165)	(0.165)	(0.170)	(0.164)
Above median: $E_t(\Pr(\text{Lose Job}_{t+12}))$		0.673^{***}	0.588^{***}	0.496^{***}	0.860^{***}
		(0.170)	(0.170)	(0.174)	(0.248)
Above median: $E_t(\Pr(\text{Leave Job}_{t+12}))$		0.977^{***}	1.075^{***}	0.751^{***}	1.077^{***}
		(0.176)	(0.176)	(0.193)	(0.177)
Above median: $E_t(\Pr(\text{Find}\mid\text{Lose}_{t+3}))$			-0.595^{***}	-0.836^{***}	-0.311
			(0.169)	(0.170)	(0.229)
$E_t(\Pr(\text{Offer}_{t+3}))$				0.019^{***}	
				(0.003)	
Above median: $E_t(\Pr(\text{Lose Job}_{t+12})) \times$ Above median: $E_t(\Pr(\text{Find}\mid\text{Lose}_{t+3}))$					-0.604^{*}
					(0.341)
$E_t(\pi_{t+12})$	-0.023	-0.024	-0.027	-0.022	-0.027
	(0.021)	(0.022)	(0.022)	(0.023)	(0.022)
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	4132	4131	4130	3922	4130
R-squared	0.125	0.142	0.145	0.158	0.146

Notes: Dependent variable is weekly job search hours (winsorized). The four core expectations from Table 1 are replaced by indicators for responses strictly above the sample median in the employed sample with non-missing search hours; coefficients therefore measure search-hour differences relative to workers at or below the median. Column (4) retains the continuous expected-offer control, and Column (5) interacts the above-median job-loss and job-finding indicators. All specifications include time and state fixed effects, are weighted by survey weights, and control for age polynomial, education, race, gender, marital status, and household income. Standard errors clustered by individual. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The indicator specification delivers the same qualitative pattern. Median search among employed workers is about 2 hours per week, so these differences are economically meaningful. Workers with above-median macro pessimism search about 0.8 hours more per week, and workers with above-median perceived job-loss risk search about 0.9 hours more per week. Above-median voluntary-separation beliefs are also associated with more search, while above-median job-finding beliefs are associated with less search in the additive specifications. The interaction column makes the precautionary interpretation clearer as among workers with below-median job-finding beliefs, moving above the median in perceived job-loss risk raises search by about 0.9 hours per week; when job-finding beliefs are also above the median, this incremental effect is smaller by about 0.60 hours. In other words, the search premium associated with high perceived separation risk is concentrated among workers who are not confident they can quickly find a new job after separation.

2.3 From macro pessimism to personal separation risk

How do aggregate expectations map into personal beliefs that are directly relevant for search? Table 3 investigates this link by regressing personal expectation variables on macro unemployment pessimism, controlling for demographics and fixed effects. The estimates reveal a sharp asymmetry: macroeconomic pessimism maps strongly into perceived separation risk but has essentially no association with perceived job-finding probability conditional on job loss.

Workers who are more pessimistic about aggregate unemployment report substantially higher perceived job-loss probability: a worker at the 75th percentile of macro pessimism perceives roughly 4 percentage points higher separation risk than a worker at the 25th percentile. Macro pessimism also correlates with voluntary separation expectations. It has essentially no relationship, however, with perceived job-finding probability conditional on job loss. The asymmetry is important: aggregate pessimism is linked primarily to perceived *inflow risk* into unemployment, not to perceived duration once unemployed.

Consistent with this pattern, controlling for personal expectations reduces the macro pessimism coefficient in the search regression by about 40 percent. A product-of-coefficients calculation attributes most of this reduction to job-loss expectations, with a secondary role for voluntary-separation beliefs. In the remainder of the paper, the quantitative model uses the job-loss-risk relationship to evaluate the aggregate implications of precautionary on-the-job search.

Table 3. Macro Expectations and Individual Labor Market Expectations

	(1) P(Lose Job)	(2) P(Leave Job)	(3) P(Find Job)
$E_t(\Delta \Pr(U_{t+12} \geq 0))$	0.1693***	0.1362***	-0.0034
	(0.0062)	(0.0080)	(0.0111)
$E_t(\pi_{t+12})$	0.0001	-0.0004	-0.0013***
	(0.0003)	(0.0003)	(0.0004)
Observations	92634	92650	92824
R-squared	0.075	0.060	0.068

Notes: Dependent variables are subjective probabilities (0–1). Regressor is macro unemployment pessimism, $\Delta \Pr(U_{t+12} \geq 0)$, with expected inflation as a control. Sample: employed workers; time and state fixed effects plus demographics; weighted by survey weights; standard errors clustered by individual. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

2.4 Do beliefs predict realized separations?

A key concern is whether reported job-loss probabilities contain information about future outcomes or instead capture noise or broad sentiment. The SCE's panel structure allows a direct check: do workers who report higher job-loss probability actually separate at higher rates?

Table 4 shows that perceived job-loss risk predicts subsequent separations. A ten percentage point increase in perceived job-loss probability is associated with a 1.3 percentage point higher realized separation rate over the subsequent three months. Although the belief horizon is 12 months, the measure forecasts near-term separations, consistent with persistent differences in separation risk across workers and jobs.[7] Appendix Table A14 shows that this relationship persists across multiple horizons (1–9 months and censored 12-month outcomes), and Appendix Table A15 shows that it remains positive and significant under both broad non-employment and strict unemployment outcome definitions.

Figure 2, Panel A, provides a nonparametric view. Sorting workers into quintiles of subjective job-loss expectations yields an upward pattern: workers in the top quintile separate at substantially higher rates than workers in the bottom quintile. This evidence is consistent with Stephens (2004) and Mueller et al. (2021), who document that subjective job-loss expectations in household surveys predict actual displacement.

The same logic has an implication for realized separations. If outside offers insure job loss, a relevant aggregate margin is the probability that a displaced worker moves into a

[7] The 12-month belief horizon exceeds the 3-month outcome window, but persistent heterogeneity in separation risk across workers implies that beliefs should predict near-term separations even if the stated horizon is longer.

Table 4. Validation: Subjective Job Loss Expectations Predict Separations (Q10-based status)

	(1) None	(2) Controls	(3) Time FE	(4) State + Time FE	(5) State×Time FE
$E_t P(\text{Lose Job}_{t+12})$	0.1547***	0.1320***	0.1314***	0.1307***	0.1290***
	(0.0129)	(0.0121)	(0.0120)	(0.0118)	(0.0107)
Controls	No	Yes	Yes	Yes	Yes
Fixed Effects	None	None	Time	State, Time	State×Time
Observations	56412	55993	55993	55832	55832
R-squared	0.021	0.049	0.056	0.059	0.205

Notes: Job-loss expectations are scaled to $[0, 1]$, so coefficients are percentage-point changes in separation probability. Status is constructed from Q10 dummies: employed/attached = FT, PT, leave; unemployed = want work, laid off; NILF = disabled, retiree, student, homemaker. Exact-horizon outcomes require observation at $t + h$ (xtset + F#.). Regressions are weighted by survey weights; SEs clustered by individual.

new job without an intervening non-employment spell. Appendix Table A16 examines this margin in the CPS Displaced Worker Supplement from 1984 to 2024, using IPUMS CPS data and a displaced-worker sample similar to the one used in Farber (2017) (Flood et al., 2025). Among displaced workers who are re-employed full time, the share reporting zero weeks without work is lower in recession displacement years, and it also falls when national unemployment rises during the displacement year. The result survives controls for displacement reason, years since displacement, and lost-job occupation and industry. I read this as evidence that the offer-conversion margin contracts in downturns. The quantitative model therefore isolates the belief-driven search-insurance channel rather than trying to match the full cyclicality of direct re-employment after displacement.

2.5 Mechanism: Evidence of a precautionary motive

The results so far show that perceived job-loss risk predicts search intensity. I next ask whether the relationship reflects a precautionary motive—searching to secure job continuity in the event of separation—rather than a generic tendency to search.

Stated search motives. The SCE Labor Market Module asks employed workers who report searching to indicate why they are looking for a new job. Table 5 uses this information to test a direct implication of precautionary search: if higher perceived job-loss risk induces search for insurance, it should raise the likelihood that workers cite *job security* as their motive.

Workers with higher perceived job-loss probability are substantially more likely to cite job security as a reason for searching. Voluntary separation expectations, by contrast, enter with the opposite sign: workers who expect to leave voluntarily are *less* likely to cite job security and more likely to cite opportunity-related motives (better pay, advancement). This difference helps separate job-loss expectations from a general desire to move jobs. Finally, conditional on personal expectations, macro pessimism has no independent association with citing job security, suggesting that the security motive operates primarily through beliefs about one's own job-loss risk.

These patterns help distinguish precautionary search from a generic propensity to search: the same beliefs that predict higher search intensity also predict that workers attribute their search to employment stability concerns.

Table 5. Direct Evidence: Why Are Employed Workers Searching?

	(1) Searching for Job Security	(2) Ladder / Upgrade Search
$E_t(\Pr(\text{Lose Job}_{t+12}))$	0.588***	-0.271***
	(0.071)	(0.068)
$E_t(\Pr(\text{Leave Job}_{t+12}))$	-0.145***	0.040
	(0.048)	(0.049)
$E_t(\Delta \Pr(U_{t+12} \geq 0))$	-0.026	0.054
	(0.061)	(0.059)
Time FE	Yes	Yes
State FE	Yes	Yes
Observations	707	707
R-squared	0.290	0.211

Notes: Column 1 equals 1 if the worker reports searching due to job stability concerns or having received a layoff notice. Column 2 equals 1 if the worker reports searching for better pay, advancement, hours, fit, quality of life, commute, or wanting a change. Expectations are scaled to $[0, 1]$. Sample: employed workers in the SCE Labor Market Module who report searching. All specifications include time and state fixed effects and demographic controls; weighted by survey weights. Standard errors clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

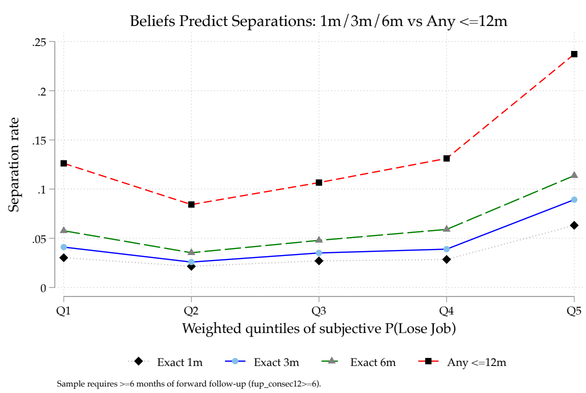
If workers search to insure against job loss, the value from searching while employed should be highest when perceived job-loss risk is high and expected re-employment prospects are weak. Table 6 tests this by allowing the effect of macro pessimism on search to vary with (i) perceived job-loss risk and (ii) perceived re-employment prospects. [8]

Two patterns stand out. First, the interaction between macro pessimism and perceived job-loss probability is positive and statistically significant. Aggregate pessimism induces a larger increase in search among workers who perceive higher personal separation risk.

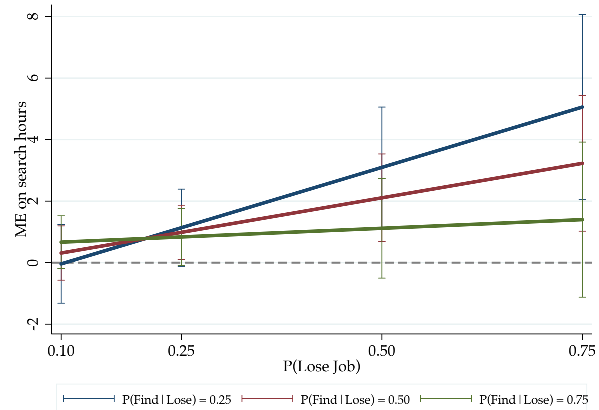
[8]Because the expectations are centered, the main coefficients describe effects at average beliefs, and the interaction terms capture how the response varies across workers.

Second, the three-way interaction between macro pessimism, perceived job-loss risk, and perceived job-finding probability is negative and statistically significant. The amplification from job-loss risk is strongest when workers are less confident about finding a job quickly. This is the case in which an outside offer provides the most insurance: high probability of needing it and high expected cost if separation occurs without one.

Figure 2, Panel B, plots the marginal effect of macro pessimism on search at different points in the joint distribution of (Pr(Lose), Pr(Find)). The effect is largest in the high-risk, low-prospect region, where workers see both a high chance of job loss and a low chance of rapid re-employment. It is smallest when separation risk is low or re-employment prospects are strong.



Panel A. Realized separations by perceived job-loss risk



Panel B. Marginal effect of macro pessimism on search

Notes: Panel A sorts employed workers into quintiles of perceived 12-month job-loss probability and plots realized separation rates at several horizons. Panel B reports marginal effects from the interaction specification in Table 6, using centered expectation variables. It shows how the effect of macro unemployment expectations on search hours varies with perceived job-loss risk and perceived job-finding probability. All models include state×time fixed effects and full controls.

Figure 2. Mechanism Evidence: Beliefs Predict Risk and Shape Search Responses

These results speak most directly to work on employed search and precautionary on-the-job search. Faberman et al. (2022) show that employed search is common and productive, while Pilossoph and Ryngaert (2024) and Raposo (2025) show that expectations can affect employed search through wage-related motives. The evidence here points to a different margin. Pessimistic workers do not search more only because they expect better wage opportunities elsewhere. They search more when they think their current job is at risk, especially when they are less confident about quick re-employment. In this sense, the results complement Ahn and Shao (2017) and Simmons (2024), who emphasize precautionary on-the-job search. The contribution here is to measure the belief margin directly: subjective job-loss probabilities predict search intensity, stated job-security motives, and

Table 6. Macro Pessimism and Expectations about Personal Outcomes

	(1) Baseline	(2) + Macro×Lose	(3) + Macro×Find	(4) + Lose×Find	(5) Three-way
$E_t(\Delta\,\Pr(U_{t+12} \geq 0))$	0.906^{**}	0.659	0.493	0.452	0.527
	(0.412)	(0.405)	(0.453)	(0.451)	(0.414)
$E_t(\Pr(\text{Lose Job}_{t+12}))$	1.841^{***}	1.616^{***}	1.645^{***}	1.694^{***}	1.592^{***}
	(0.502)	(0.498)	(0.496)	(0.499)	(0.500)
$E_t(\Pr(\text{Find Job}_{t+3}))$	-1.259^{***}	-1.235^{***}	-1.220^{***}	-1.150^{***}	-1.057^{***}
	(0.352)	(0.352)	(0.352)	(0.358)	(0.360)
$E_t(\Delta\,\Pr(U_{t+12} \geq 0)) \times E_t(\Pr(\text{Lose Job}_{t+12}))$		3.892^{**}	3.552^{*}	3.445^{*}	3.817^{**}
		(1.794)	(1.917)	(1.917)	(1.772)
$E_t(\Delta\,\Pr(U_{t+12} \geq 0)) \times E_t(\Pr(\text{Find Job}_{t+3}))$			-0.224	0.036	0.909
			(1.335)	(1.340)	(1.259)
$E_t(\Pr(\text{Lose Job}_{t+12})) \times E_t(\Pr(\text{Find Job}_{t+3}))$				-1.364	-0.794
				(1.456)	(1.442)
$E_t(\Delta\,\Pr(U_{t+12} \geq 0)) \times E_t(\Pr(\text{Lose Job}_{t+12})) \times E_t(\Pr(\text{Find Job}_{t+3}))$					-13.438^{**}
					(5.442)
$E_t(\pi_{t+12})$	-0.023	-0.024	-0.023	-0.024	-0.023
	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)
F-stat		4.71			6.10
p-value		0.030			0.014
Observations	4130	4130	4130	4130	4130
Adj. R-squared	0.177	0.179	0.179	0.179	0.181

Notes: Standard errors in parentheses. Dependent variable is weekly job search hours (winsorized). All specifications include state×time fixed effects and full controls. Standard errors are clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

later separations in the same survey data.

2.6 Robustness

The baseline estimates document a cross-sectional association between expectations and search. One concern is omitted variables: workers in declining industries or weak local labor markets may report both high pessimism and high search intensity because both reflect underlying conditions rather than a behavioral response. The robustness exercises address this and related concerns. Full tables appear in Appendix B.

The main relationship is not driven by the COVID-19 months in the sample, and the inference is similar when standard errors are two-way clustered by worker and survey month. Excluding April–December 2020 leaves the preferred job-loss coefficient slightly larger than in the baseline, and the coefficient remains precisely estimated under two-way clustered standard errors.

Alternative fixed effects. Replacing separate state and time fixed effects with state×time interactions absorbs all local aggregate variation, so identification relies on differences across workers within the same state-month cell. Under this specification, the macro pessimism coefficient declines modestly but remains statistically significant, and the job-loss expectations coefficient is essentially unchanged (Table A2). With industry×time fixed effects, which absorb sector-level cyclical conditions, the results are similar (Table A3). These specifications reduce the scope for local or sectoral shocks to explain the estimates.

Functional form and margins. Using alternative transformations of search hours ($\log(1 + y)$ and an arc-percent transformation) and Poisson specifications leaves the qualitative results intact (Table A9). The arc-percent specification follows a DHS-style cross-sectional scaling, $(y - \bar{y})/(0.5(y + \bar{y}))$, which avoids relying on the scale-dependent inverse-hyperbolic-sine transformation. At the extensive margin (Table A10), personal separation expectations predict whether a worker searches at all, whereas macro pessimism does not once personal expectations are included. Macro beliefs primarily shift intensity among those already searching, while personal job-loss beliefs operate along both margins.

Within-person variation. First-difference and individual fixed effects specifications exploit within-worker variation over time (Table A11). The job-loss coefficient remains positive but becomes imprecise, reflecting the short panel dimension: workers appear in the LMM only about 1.4 times on average. Voluntary-separation beliefs remain predictive in some within-person specifications, but this margin is ancillary to the precautionary interpretation emphasized here.

Employed versus unemployed workers. Table A12 compares the role of expectations by employment status. Among employed workers, macro pessimism predicts search intensity. Among the unemployed, the relationship is essentially zero. This asymmetry supports the paper's focus on the employed margin: expectations matter for workers who have a job to insure, not for those already unemployed and searching by necessity.

2.7 External validation in the ECB Consumer Expectations Survey

The SCE is well suited to the main analysis because it combines aggregate expectations, personal labor-market expectations, detailed search hours, stated search motives, and panel outcomes for the same employed workers. The ECB Consumer Expectations Survey provides a useful external check on the same link in a different setting.[9]

[9] I use the ECB CES microdata release dated 28 April 2026, covering the regular monthly and quarterly modules through 2026Q1. The regular quarterly module measures job-loss risk over a 3-month horizon and contains search outcomes such as active search, the probability of looking for a job, and search intensity. The analysis uses the 11 euro area countries in the release. All specifications include survey-round and

Appendix Table A17 reports three exercises using employed respondents in the ECB labor-market module matched to monthly expectations. The first two reproduce the main links behind the SCE mechanism. First, macroeconomic pessimism predicts higher perceived job-loss risk. After controlling for other macro and household expectations, a 10 percentage point higher expected unemployment rate is associated with about a 0.9 percentage point higher perceived probability of losing the current job over the next 3 months. Second, perceived job-loss risk predicts search. A 10 percentage point higher perceived job-loss probability is associated with a 4.7 percentage point higher probability of actively looking for a job and a 6.2 percentage point higher reported probability of looking for a job over the next 3 months. Search intensity also rises with perceived job-loss risk.

The ECB evidence supports the external validity of the main link: workers who expect worse aggregate labor-market conditions perceive higher own job-loss risk, and workers who perceive higher job-loss risk search more. The ECB is less suited to the richer heterogeneity tests in the SCE. In the regular quarterly module, the core job-finding probability is observed for unemployed respondents, not employed respondents. The topical labor-market module contains an employed-worker analogue, expected time to find an equivalent job, but it is measured in a separate module rather than in the same quarterly interview. I use it only as a supplementary predetermined-belief check, merging the round-41 topical response forward to later quarterly observations and, more tightly, to the nearest subsequent quarterly round. This exercise does not replicate the SCE heterogeneity result: in the ECB data, the search response to job-loss risk is not stronger among workers who expect it would take longer to find an equivalent job. I therefore use the ECB as evidence that the belief-to-risk-to-search chain is not unique to the SCE, while the direct evidence on precautionary motives and heterogeneity by job-finding beliefs comes from the SCE.

The estimates in this section are associations, and identifying the causal effect of expectations on search would require variation that moves what workers believe about the labor market without moving their search incentives through some other channel, and such variation is hard to come by: the aggregate shocks that shift beliefs also shift labor-market conditions, and the personal shocks that shift beliefs often shift search directly. What the cross-sectional design delivers instead is a comparison of workers in the same time and place who happen to disagree about the economy. That comparison cannot rule out unobserved worker or job heterogeneity, but three features of the data make a

---

country fixed effects, demographic controls, ECB CES weights, and respondent-clustered standard errors. The controlled specifications additionally include expected inflation, expected economic growth, current perceived unemployment, and expected household financial conditions. Expected unemployment is also included as a control in the search regressions, where it is not the focal regressor.

pure omitted-variables story uncomfortable: the beliefs predict realized separations, they predict security-related search motives, and they matter most exactly where an outside offer is most valuable.

It is still worth asking what makes workers in the same state and month disagree about the macroeconomy in the first place. The 2016 presidential election offers one clean illustration. Following Coibion et al. (2020) and Mian et al. (2018), I compare workers in Republican- and Democrat-won states (dropping swing states) around November 2016: partisan priors interacting with the election outcome opened up a roughly 6 percentage point gap in macro unemployment expectations across otherwise similar local labor markets, with flat pre-trends, and the gap did *not* carry over to personal perceived job-loss risk—workers separate their views about the aggregate economy from the read on their own job security, the same asymmetry documented in Section 2.3. The implied search response has the expected sign but is imprecise at the quarterly frequency of the module. I read the episode as an illustration of where within-cell belief dispersion comes from rather than as a stand-alone identification strategy, and report the full event-study and difference-in-differences estimates in Appendix B.7.

Overall, expectations about the aggregate economy are tied to how hard employed workers search, and the link runs mainly through perceived job-loss risk. The aggregate-pessimism coefficient survives controlling for personal beliefs, suggesting that it still captures perceived risk beyond what the SCE's personal-expectation questions span. The model in the next section formalizes this link and traces its aggregate implications.

3 Model

The empirical analysis documented three facts about employed workers in the Survey of Consumer Expectations. Workers who report higher job-loss risk search more while employed. This response is stronger when expected re-employment prospects are weak and is reflected in job-security search motives. Reported job-loss probabilities also predict subsequent separations.

This section develops a Diamond–Mortensen–Pissarides search-and-matching model with on-the-job search. The model has two main ingredients: separation risk rises in recessions, and employed workers choose search effort before a possible job loss. When a separation occurs, a worker with an outside offer moves directly into a new job next month, while a worker without an offer enters unemployment. This is the mechanism emphasized by Ahn and Shao (2017): on-the-job search is valuable not only because it may improve the current job ladder, but also because it creates an option that is useful when the worker faces job loss. Here, the search decision is disciplined by workers' reported job-loss beliefs, and the job-ladder extension makes the same offer technology account for ordinary employer-to-employer mobility. Search affects unemployment dynamics by changing the fraction of separations that become unemployment spells.

The baseline model abstracts from voluntary job-to-job transitions absent separation, heterogeneous match quality, and endogenous separations. These simplifications keep the focus on the response of employed search to perceived job-loss risk. I then add a job-ladder extension. The extension uses the same offer technology but gives outside options two possible uses. If the current job survives, an offer can generate an ordinary job-ladder move. If the current job is destroyed, the outside option can insure the separation. Matching total E-to-E mobility and the share of E-to-E moves related to job loss pins down how much of the offer portfolio is usable at separation. The baseline shows the mechanism in its cleanest form; the extension asks how much remains once ordinary job-ladder mobility is also matched.

Section 4 describes the calibration, and Section 5 uses the calibrated model to quantify the aggregate implications of precautionary on-the-job search.

3.1 Environment

Time is discrete and indexed by $t = 0, 1, 2, \ldots$, with each period interpreted as one month. The economy is populated by a unit mass of risk-neutral workers and a continuum of risk-neutral firms. Workers discount future payoffs at factor $\beta \in (0, 1)$. Firms post vacancies at flow cost $\kappa > 0$ and enter freely.

Aggregate states and separations. The aggregate state is $z \in \{G, B\}$, where G denotes an expansion and B a recession. Productivity is $y(z)$, and the state follows a Markov transition matrix $\Pi(z, z')$. An employed worker's current match is destroyed during the month with probability $\delta(z)$. Separation risk is higher in recessions, $\delta(B) > \delta(G)$.

Beliefs. Workers may perceive separation risk differently from the objective probability. Let $\tilde{\delta}(z)$ denote the worker's perceived monthly probability of job loss. In the calibration, a level parameter ζ matches the average level of perceived job-loss risk and a state component $m(z)$ allows beliefs to move differently across aggregate states.[10] Beliefs affect search choices

[10] The exogenous separation block is a reduced-form representation of the job-loss risk workers report in the survey, not a theory of why matches end. In a richer model, match-specific productivity, firm distress, or worker-firm sorting would generate heterogeneous and partly endogenous separation probabilities. Those forces would change the source of $\delta(z)$, but the employed worker's search problem would still depend on the perceived probability of job loss and the value of having an outside offer when that loss occurs. I therefore

and worker continuation values. Realized worker flows are governed by the objective separation probability $\delta(z)$.

Timing. Within a period t, the realized aggregate state is z_t. Firms post vacancies and matching occurs. Employed workers choose search intensity $s(z_t)$ and may receive outside offers. Separations are realized at the end of the month. Matches formed during month t become active at $t+1$ and are therefore not exposed to an additional separation draw within their month of formation. The economy then transitions to z_{t+1} according to Π, and continuation values are evaluated in z_{t+1}.[11] This timing is convenient empirically: it matches the monthly frequency of the SCE and the notion that current labor-market conditions govern offers and separations over the month, while expectations about the future state matter through discounted continuation values.

Matching technology Let u denote unemployment, v vacancies, and $e = 1-u$ employment at the beginning of the month. Unemployed workers supply one unit of search. Employed workers supply effective search units $g(s(z))$, where $g(s) = \log(1+s)$. Total search units are

$$S(z) = u(z) + e(z)g(s(z)).$$

Following the searcher-pool formulation in Mukoyama et al. (2018), market tightness is vacancies per search unit,

$$\theta(z) = \frac{v(z)}{S(z)}.$$

The matching function is

$$M(S,v) = \chi S^{\alpha} v^{1-\alpha}, \qquad \alpha \in (0,1),\ \chi > 0.$$

Tightness summarizes congestion in the pool of searchers: when θ rises, searchers are more likely to meet firms, while vacancies are harder to fill. The monthly job-finding probability for an unemployed worker in state z is

$$p_u(z) = 1 - \exp\big[-\chi\theta(z)^{1-\alpha}\big], \tag{2}$$

use $\tilde{\delta}(z)$ as the state variable pinned down by the SCE beliefs and keep the destruction process outside the model. This choice isolates the precautionary-search margin, while leaving endogenous quits, match-quality selection, and the effect of search on match destruction for richer extensions.

[11]Thus, within-month transition probabilities (job finding, offer arrival, separations) are functions of the *current* state z_t, while Π governs only the distribution of next month's continuation values.

and the probability a vacancy is filled in state z is

$$p_f(z) = 1 - \exp\big[-\chi\theta(z)^{-\alpha}\big]. \tag{3}$$

3.2 Employed Search and Offers

An employed worker chooses search intensity $s \geq 0$ at flow cost

$$c(s) = \frac{\gamma}{2}s^2.$$

Search produces an outside offer with probability

$$p_{\text{offer}}(z,s) = 1 - \exp\{-\eta\chi\theta(z)^{1-\alpha}\psi g(s)\}, \qquad g(s) = \log(1+s).$$

Its derivative with respect to search effort is

$$p_{\text{offer},s}(z,s) = \exp\{-\eta\chi\theta(z)^{1-\alpha}\psi g(s)\}\eta\chi\theta(z)^{1-\alpha}\psi g'(s). \tag{4}$$

The parameter ψ governs the effectiveness of employed search effort and γ governs the cost of search. The parameter η allows employed search to be less effective than unemployed search per unit of effort.[12]

The derivative $p_{\text{offer},s}(z,s)$ is the marginal return to employed search effort.[13] The precautionary motive does not require this return to rise in recessions. Beliefs matter by raising the value of search as insurance against job loss.

The same offer technology can serve two roles. If the current job survives, an offer may generate an ordinary job-ladder move. If the current job is destroyed, an offer may allow the worker to avoid unemployment. Let ρ_J be the probability that an offer is acceptable as a voluntary job-ladder move conditional on no separation. Let ρ_I be the probability that an offer is usable as insurance conditional on separation. The first parameter pins down ordinary E-to-E mobility; the second pins down how much of the offer portfolio protects workers at separation. The baseline outside-option model is the special case $\rho_J = 0$ and $\rho_I = 1$.

The model uses a parsimonious reduced-form value for ladder moves. In the main

[12]Only the product $\eta\psi$ is identified from the offer-probability target, because η and ψ enter the offer hazard multiplicatively. Setting $\eta = 0.10$ is therefore a normalization that reflects the lower contact intensity of employed search, and ψ absorbs the remaining scale.

[13]Graves and Huckfeldt (2026) make a related distinction between active search, in which workers deliberately apply or contact employers, and passive search, in which jobs arrive through referrals or other unsolicited channels. Their evidence is useful for separating search effort from the return to that effort.

specification, an acceptable ladder offer gives a gain proportional to the employment surplus:

$$\Delta_J(z') = \omega_J\big(W(z') - U(z')\big).$$

This keeps the ladder value tied to the same Bellman objects that determine the insurance value of avoiding unemployment. The parameter ω_J is calibrated from the average wage gain of E-to-E movers.

3.3 Worker Values

Let $W(z)$ and $U(z)$ denote the worker's continuation values of employment and unemployment, evaluated at the worker's search policy and perceived separation risk $\tilde{\delta}(z)$. To keep the notation light, write $p_e(z) \equiv p_{\text{offer}}(z, s(z))$. The value of employment averages over two events. If the worker perceives the match to survive, a usable offer may generate a ladder gain. If the worker perceives the match to be destroyed, a usable offer keeps the worker employed next period; otherwise the worker enters unemployment:

$$\begin{aligned} W(z) = w(z) - c(s(z)) + \beta \sum_{z'} \Pi(z, z')\Big\{&(1 - \tilde{\delta}(z))\big[W(z') + \rho_J p_e(z)\Delta_J(z')\big] \\ &+ \tilde{\delta}(z)\big[\rho_I p_e(z)W(z') + (1 - \rho_I p_e(z))U(z')\big]\Big\}. \end{aligned} \tag{5}$$

The unemployed value is

$$U(z) = b + \beta \sum_{z'} \Pi(z, z')\big[p_u(z)W(z') + (1 - p_u(z))U(z')\big].$$

The first-order condition for employed search is

$$c'(s(z)) = \beta p_{\text{offer},s}(z, s(z)) \sum_{z'} \Pi(z, z') \left[(1 - \tilde{\delta}(z))\rho_J \Delta_J(z') + \tilde{\delta}(z)\rho_I\big(W(z') - U(z')\big)\right]. \tag{6}$$

The bracketed term separates the two motives for search. The first term is the ladder motive: an offer is valuable because it may improve the worker's job if the current match survives. The second is the precautionary motive: an offer is valuable because it may prevent unemployment if the current match is destroyed. A higher perceived separation probability raises the value of search as insurance against job loss. Search rises with perceived risk when this insurance value is large enough relative to the ladder value displaced by the higher chance of separation.

3.4 Flows

Realized flows are governed by objective separation risk. The employment-to-unemployment probability is

$$p_{EU}(z) = \delta(z)\left[1 - \rho_I p_{\text{offer}}(z, s(z))\right]. \tag{7}$$

Search lowers unemployment inflows only through the insurance component. Voluntary job-ladder moves change jobs but do not directly prevent unemployment.

The model generates two E-to-E flows:

$$p^J_{EE}(z) = \big(1 - \delta(z)\big)\rho_J p_{\text{offer}}(z, s(z)), \tag{8}$$

and

$$p^I_{EE}(z) = \delta(z)\rho_I p_{\text{offer}}(z, s(z)). \tag{9}$$

Total E-to-E mobility is $p^J_{EE}(z) + p^I_{EE}(z)$. The ratio

$$\frac{p^I_{EE}(z)}{p^J_{EE}(z) + p^I_{EE}(z)}$$

maps the model to the observed share of E-to-E moves that are associated with job loss or anticipated layoff. This moment is informative about ρ_I: self-reported job-loss E-to-E transitions are a conservative measure of insured separations, because workers who anticipate job loss and move successfully may describe the move as a better-job transition rather than a layoff.

3.5 Firm Values and Equilibrium

Ladder moves generate quit risk for firms. A filled job survives to the next month only if the match is not destroyed and the worker does not leave for an acceptable ladder offer. The firm value is

$$J(z) = y(z) - w(z) + \beta(1 - \delta(z)) \sum_{z'} \Pi(z, z') \left[\big(1 - \rho_J p_{\text{offer}}(z, s(z))\big) J(z')\right].$$

The insurance parameter ρ_I does not enter firm survival because an insured separation uses an offer only after the current match has already been destroyed. Free entry sets the cost of a vacancy equal to the expected value of a filled job.

Wages are determined by Nash bargaining over the full equilibrium match values, not by the closed-form DMP wage that would apply without on-the-job search.[14] Let the worker's match surplus be

$$X(z) = W(z) - U(z).$$

Taking the search policy, offer probability, and quit risk as part of the equilibrium, the wage in state z solves

$$\max_{w(z)} X(z)^{\phi} J(z)^{1-\phi}.$$

Since the wage is a transfer between the worker and firm, the Nash condition is

$$(1-\phi)X(z) = \phi J(z). \tag{10}$$

Perceived separation risk enters the worker surplus through both the Bellman value and the search policy. Firm values and realized flows continue to use objective separation risk. Thus beliefs can affect wages through the worker side of the Nash bargain, while the firm's destruction process is governed by the actual law of motion for jobs.

Equation (10) is solved jointly with the worker value, firm value, free-entry, and search equations. In the job-ladder extension, $X(z)$ includes the worker's ladder and insurance value, while $J(z)$ includes the quit-survival term $(1 - \rho_J p_{\text{offer}}(z, s(z)))$. An equilibrium consists of worker values, firm values, wages, tightness, employed search, and flows such that employed search satisfies (6), firms satisfy free entry, wages satisfy (10), and stocks obey the law of motion implied by p_{EU} and p_u.

3.6 Analytical Implications

Equation (6) is the key margin. Search has a ladder value and an insurance value. Higher perceived job-loss risk shifts weight toward the insurance branch. Whether search rises depends on which branch is more valuable at the margin.

Assumption 1 (Regularity)**.** *The employed-search choice is interior, $p_{offer,s}(z,s) > 0$, $c''(s) > 0$, and the offer technology satisfies $g'(s) > 0$ and $g''(s) \le 0$, so that $p_{offer,ss}(z,s) < 0$.*

Proposition 1 (Perceived risk raises search when the insurance value is high)**.** *Holding tightness, continuation values, and ladder values fixed, employed search is increasing in perceived*

[14]The bargaining protocol treats outside offers as mobility options rather than triggers for renegotiating the incumbent match. This is the Burdett–Mortensen-style fork rather than a Postel-Vinay–Robin counteroffer environment. The worker's ladder value is part of the private continuation value entering $X(z)$, while the incumbent firm faces quit risk through $J(z)$.

separation risk whenever

$$\sum_{z'} \Pi(z,z')\rho_I\big(W(z') - U(z')\big) > \sum_{z'} \Pi(z,z')\rho_J\Delta_J(z').$$

This is the model counterpart of the empirical result. When workers think job loss is more likely, an offer is less about improving a surviving match and more about avoiding unemployment. If that insurance value is large enough, perceived risk raises search.

Proposition 2 (Search lowers unemployment inflows when offers insure separations)**.** *If $\rho_I > 0$ and $p_{offer,s}(z,s) > 0$, then $p_{EU}(z)$ is decreasing in employed search effort.*

The sign is immediate from (7). More search raises the probability of having an offer in hand; if some offers can be used at separation, fewer separations become unemployment spells.

Proposition 3 (A sufficient condition for countercyclical search)**.** *Search is higher in recessions than expansions if the recession-state marginal benefit of search in* (6) *is at least as large as the expansion-state marginal benefit at the relevant effort levels.*

This condition makes clear why countercyclical search is not automatic. Recessions lower tightness, which makes effort less effective. Search rises only if the higher perceived-risk motive is strong enough to offset that force and any changes in surplus or ladder value. This is exactly what the counterfactuals below test: when perceived job-loss probabilities are held fixed across states, the precautionary motive loses its cyclical force and search no longer rises in the same way.

Proposition 4 (Precautionary search dampens recession unemployment)**.** *For a given recession equilibrium, if endogenous search exceeds the fixed-search level and $\rho_I > 0$, then unemployment is weakly lower than in the fixed-search counterfactual.*

The proposition pins down the sign, not the size. The size depends on how many offers are actually usable when a job ends. That is why the calibration separates ordinary ladder mobility from insurance against job loss. Once that distinction is imposed, beliefs still matter for worker behavior, but the aggregate unemployment effect is disciplined by the share of offers that can prevent unemployment.

4 Calibration

The calibration uses two kinds of evidence. Aggregate stock-flow moments pin down the unemployment accounting. SCE moments pin down beliefs, offer receipt, and the response of search to perceived job-loss risk. I first report the baseline outside-option model. I then report the job-ladder extension, which adds E-to-E mobility and the job-loss-related share of those moves as targets. The extension asks how much of the baseline mechanism remains after ordinary job-ladder mobility is also matched.

4.1 Targets

Time is monthly. I set standard parameters outside the calibration: $\beta = 0.9967$, $b = 0.4$, $\alpha = 0.5$, $\phi = 0.5$, and an employed/unemployed contact ratio $\eta = 0.10$. Productivity takes two values, $y(G) = 1$ and $y(B) = 0.95$, with state persistence chosen to match NBER business-cycle durations. Appendix Table A19 reports the full list and sources.

The main calibration uses the SCE profile. It matches mean perceived 12-month job-loss risk in the search sample, 0.212; the employed offer probability, 0.330; and the empirical semi-elasticity of search with respect to perceived job-loss risk, $\beta_{\text{plose}} = 1.688$. In the job-ladder extension, it also matches total monthly E-to-E mobility, 0.0385, and the share of E-to-E moves reported as job-loss or anticipated-layoff moves, 0.094.

These last two moments separate ordinary ladder moves from offers that insure job loss. Total E-to-E mobility identifies ρ_J, the probability that an offer is acceptable as a ladder move. The job-loss-related E-to-E share identifies ρ_I, the probability that an offer is usable at separation. I treat the latter moment as conservative. A worker who sees a layoff coming, searches, and moves directly to a better job may report the move as a better-job transition rather than as job loss. For this reason, I also report a calibration with higher insurance use, $\rho_I = 0.75$.

This arithmetic gives a useful upper-bound intuition before any model result. Total E-to-E mobility is 3.85 percent per month, and 9.4 percent of those moves are reported as job-loss or anticipated-layoff moves. The directly observed insured-transition component is therefore about $0.0385 \times 0.094 = 0.0036$, or 0.36 percent of employment per month. Cyclical movement in this flow is the margin through which re-optimized employed search can reduce unemployment. This composition discipline is why the job-ladder extension produces a much smaller unemployment effect than the baseline outside-option model, where all offers are usable at separation by construction.

The ladder value parameter ω_J is calibrated from the average wage gain of E-to-E movers. In the headline specification, the gain is proportional to the employment surplus, $\Delta_J = \omega_J(W - U)$. Concretely, ω_J solves

$$\omega_J \, \overline{(W - U)} = g_{EE} \, \bar{w},$$

where g_{EE} is the average proportional wage gain of E-to-E movers and $\bar{w}$ and $\overline{(W-U)}$ are ergodic means. Thus ω_J is the ladder-value parameter; Δ_J is not estimated as a separate level.

Stock-flow discipline. The unemployment stock and the U-to-E probability cannot be combined with the raw SCE E-to-U transition as three independent targets. In a two-state steady state, $u = 0.0573$ and $p_{UE} \simeq 0.28$ imply

$$p_{EU} = \frac{u p_{UE}}{1-u} \simeq 0.0170. \tag{11}$$

The raw SCE next-wave E-to-U probability is lower, about 0.0116, reflecting time aggregation, survey rotation, and movements through inactivity. I therefore target the unemployment stock and the U-to-E probability, and report the implied E-to-U probability as the stock-flow object. The objective job-loss probability is recalibrated after each update of ρ_I, because the effective E-to-U probability is

$$p_{EU}(z) = \delta(z)\left[1 - \rho_I p_{\text{offer}}(z, s(z))\right].$$

This ordering matters. Lowering ρ_I reduces the fraction of separations covered by offers. To keep the stock-flow E-to-U probability fixed, the underlying job-loss probability must then rise.

4.2 Calibration Results

The baseline outside-option calibration sets $\rho_I = 1$ and $\rho_J = 0$ by construction. It matches the same perceived-risk, offer-probability, and search-response anchors. The calibrated belief shifter is $\zeta = 0.775$, search cost curvature is $\gamma = 0.0248$, and search effectiveness is $\psi = 27.11$. In equilibrium, search rises from 0.537 in expansions to 0.678 in recessions, while the employed offer probability rises from 0.330 to 0.363.

Table 7 reports the targeted moments and calibrated parameters for the job-ladder extension. The model matches the micro anchors: mean perceived job-loss risk is 0.212, the employed offer probability is 0.330, and the model-implied search semi-elasticity is 1.688. It also matches total E-to-E mobility and the job-loss-related E-to-E share. Appendix Table A20 reports untargeted cyclical moments.

Interpretation. The calibrated $\rho_I = 0.532$ implies that slightly more than half of offer opportunities can serve as insurance against job loss. I do not read this as the literal

Table 7. Calibration Targets and Parameter Values: Job-Ladder Extension

Panel A. Calibration Targets

Moment	Data	Model	Role
Unemployment rate u	0.057	0.054	stock-flow target
Monthly U-to-E probability p_{UE}	0.280	0.280	stock-flow target
Stock-flow E-to-U probability p_{EU}	0.017	0.017	implied by u and p_{UE}
Mean perceived job-loss probability, 12 months	0.212	0.212	target
Employed offer probability	0.330	0.330	target
Search semi-elasticity β_{plose}	1.688	1.688	target
Monthly total E-to-E probability	0.0385	0.0383	target
Job-loss-related share of E-to-E	0.094	0.095	target

Panel B. Calibrated Parameters

Parameter	Description	Value
χ	Matching efficiency	0.326
κ	Vacancy cost	0.539
ψ	Search effectiveness	8.746
γ	Search cost curvature	0.000691
ζ	Belief level shifter	0.956
ρ_I	Offer usable as insurance	0.532
ρ_J	Offer acceptable as ladder move	0.108
ω_J	Ladder gain as share of surplus	0.145

Notes: The model uses the proportional-surplus ladder specification, $\Delta_J = \omega_J(W - U)$. The stock-flow E-to-U probability is implied by the unemployment and U-to-E targets rather than independently targeted from raw SCE transitions.

share of workers who receive clean advance notice of a layoff. It is the model's mapping from reported job-loss E-to-E moves into the share of offers that prevent separations from becoming unemployment spells. The calibrated $\rho_J = 0.108$ lets the model match total E-to-E flows without treating all mobility as precautionary insurance. Once this ladder margin is included, most employed search is no longer unemployment insurance. The precautionary component still matters for worker behavior, because it makes search respond to perceived job-loss risk, but it is not the dominant source of E-to-E mobility. As a level benchmark, I also re-solve the model with the insurance use of offers shut off. In the SCE calibration, recession unemployment rises by 1.67 percentage points; the effect is larger in the high-insurance row and smaller in the CPS row (Appendix Table A24). I also report an advance-information upper bound, motivated by WARN-style notice policies, in which workers with advance notice have a longer window to secure an offer before separation (Appendix Table A25). The re-optimization exercises below ask a narrower question: how much lower is recession unemployment because workers raise search when perceived risk rises? Appendix Tables A21 and A29 report robustness profiles and sensitivity checks.

5 Quantitative Results

The model delivers two quantitative results. First, pessimistic beliefs raise employed search because they make an outside offer more valuable as insurance against job loss, as in Proposition 1. Second, this precautionary motive helps explain why employed search can rise in recessions, even when vacancies are scarce and offers are harder to get. The aggregate unemployment effect is more limited once the model also matches ordinary job-to-job mobility. Beliefs matter for worker behavior; they matter for unemployment only when outside offers can be used at separation.

5.1 Search Cyclicality

Table 8 reports search and offer probabilities in expansions and recessions, the quantitative counterpart of Proposition 3. In the baseline model, search rises from 0.6 in expansions to 0.7 in recessions, an increase of 26 percent. This is not because the calibration targets search separately in the two states. The calibration targets the worker-level response of search to perceived job-loss risk. In a recession, workers search more because the insurance value of an outside offer rises.

This goes against the usual tightness argument. In recessions, vacancies are scarce relative to searchers, so offers are harder to get. If that were the only force, employed search would fall. In the calibrated model, higher perceived job-loss risk leads to the precautionary motive to dominate and search rises.[15]

The point is that a recession does more than lower the return to effort by reducing tightness. It also raises the value of avoiding unemployment. In the calibrated model, the second force is strong enough to dominate. Expectations about the aggregate economy can therefore generate countercyclical search: workers search harder precisely when outside offers are harder to obtain.

The lower rows of Table 8 repeat the comparison in the job-ladder extension. Not every E-to-E move is insurance against unemployment. Many are ordinary ladder moves, and the extension separates these two uses of outside offers. Even after doing so, search is higher in the recession state, but the offer probability is roughly flat because lower tightness offsets much of the extra effort.[16] Quantitatively, the job-ladder extension is much closer to the held-out SCE search-cyclicality moment than the baseline: employed search rises 6.4 percent in the extension, compared with a simple high-slack/low-slack SCE mean split of 4.4 percent reported in Appendix Table A20.

Table 8. Search and Offer Probabilities Across States

Profile	$s(G)$	$s(B)$	Change in s	$p_{\text{offer}}(G)$	$p_{\text{offer}}(B)$
Baseline outside-option model	0.537	0.678	26.3%	0.330	0.363
Job-ladder extension, SCE core	2.990	3.182	6.4%	0.330	0.322
Higher insurance use, $\rho_I = 0.75$	2.043	2.235	9.4%	0.330	0.327
CPS	1.959	2.152	9.9%	0.330	0.328

Notes: Each row is recalibrated, except that the higher-insurance-use case fixes $\rho_I = 0.75$. Search is in model effort units. Units are calibration-specific, so only within-row comparisons between states are meaningful. The calibration targets the response of search to perceived job-loss risk, not the level of search in each state.

5.2 Aggregate Recession Counterfactual

Table 9 reports the main recession counterfactual, which quantifies the dampening result in Proposition 4. It asks how much aggregate unemployment would increase if workers did not raise search in recessions. I keep the economy in the recession state, but force

[15] Search effort and offer probabilities need not move one-for-one, because tightness in recession worsens. In the baseline model, higher recession search raises the offer probability from 0.330 to 0.363. In the job-ladder extension, lower tightness offsets more of the increase in effort, so the offer probability is roughly unchanged.

[16] The robustness rows show that the search result is not driven by one aggressive reading of E-to-E moves. When a larger share of offers can be used at separation, search still rises in recessions. In the CPS profile, less E-to-E mobility is assigned to the insurance channel, but search also remains higher in the recession state. The behavioral result is stable: when perceived job-loss risk rises, workers search more.

employed search to stay at its expansion level. I then compare recession unemployment under fixed search with recession unemployment when workers choose search optimally.

The first row of Table 9 shows the baseline outside-option model. Unemployment rises from 5.24 percent in expansions to 8.42 percent in recessions. If employed search is held fixed at its expansion level, recession unemployment rises to 9.03 percent. The difference is 0.61 percentage points, or about 16 percent of the fixed-search recessionary increase. In this benchmark, precautionary search is an economically meaningful stabilizer: workers' extra search offsets roughly one sixth of the recessionary rise in unemployment.

Table 9. Recession Steady States: Baseline vs. Fixed Search

Profile	ρ_I	ρ_J	$u(G)$	$u(B)$	$u(B)$ fixed s	Dampening
Baseline outside-option model	1.000	0.000	5.24%	8.42%	9.03%	0.61 pp (16.0%)
Job-ladder extension, SCE core	0.532	0.108	5.39%	9.10%	9.15%	0.046 pp (1.24%)
Higher insurance use	0.750	0.102	5.39%	9.08%	9.20%	0.115 pp (3.03%)
CPS	0.223	0.079	5.40%	9.07%	9.10%	0.030 pp (0.80%)

Notes: Dampening is $100 \times [u_B^{\text{fixed search}} - u_B^{\text{reoptimized}}]$. The fixed-search path holds recession employed search at the expansion equilibrium level while using recession-state contact probabilities.

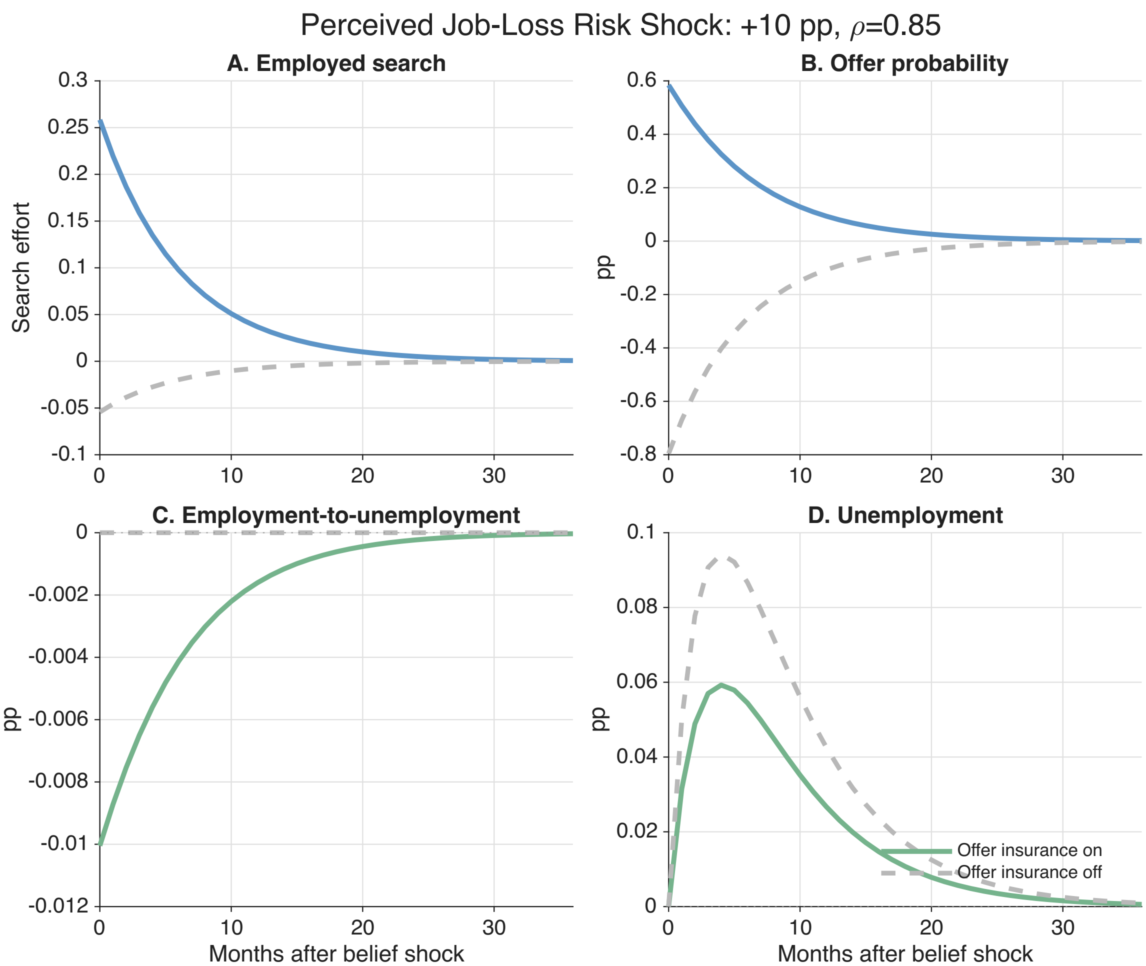
The remaining rows of Table 9 show how the job-ladder extension changes the conclusion. In the SCE core calibration, precautionary search lowers recession unemployment by 0.046 percentage points. In the CPS calibration the effect is 0.030 percentage points, and when more offers can be used at separation it rises to 0.115 percentage points. The high-insurance row is an upper reading, not the headline estimate. This is the force in Proposition 2: search lowers unemployment inflows only through offers that are usable at separation. Once job-to-job moves are matched separately, only part of the offer portfolio is available as insurance against job loss. Search still responds to beliefs, but only a small share of the extra offers prevents unemployment.

The baseline model shows that the mechanism can matter for aggregates if outside offers mainly insure job loss. The job-ladder extension disciplines that channel using E-to-E mobility. Much of employed search is ordinary mobility rather than unemployment insurance. The behavioral result survives, but the aggregate unemployment effect becomes modest. Beliefs matter for what workers do; they matter for unemployment only to the extent that the additional search produces offers that are usable at separation.

Figure 3 reports the belief-shock exercise, which isolates the search decision from changes in actual labor-market conditions. This exercise is the cleanest quantitative version of Proposition 1. The aggregate state, objective separation risk, productivity, and ladder parameters are held fixed. I then raise perceived 12-month job-loss risk by 10 percentage points and let the belief shock fade with monthly persistence $\rho = 0.85$.

The first panels of Figure 3 show the behavioral response. Workers search more because an outside offer is more valuable when they think job loss is more likely. At impact, perceived 12-month job-loss risk rises from 31 percent to 41 percent, search rises by about 8 percent, and the offer probability rises by 0.58 percentage points. This isolates the belief mechanism from recession fundamentals: even with actual separation risk and productivity fixed, perceived risk alone raises search.

The E-to-U probability falls by only 0.010 percentage points, because only some offers can be used at separation. This small aggregate response is informative about where the channel operates. Beliefs first move worker behavior. They affect unemployment only through the narrower channel of offers that arrive before, and can be used at, job loss.



Notes: The figure reports responses in the calibrated job-ladder extension to a 10 percentage point increase in perceived 12-month job-loss risk with monthly persistence $\rho = 0.85$. Actual labor-market conditions are held fixed. The solid line keeps the offer-insurance channel active. The dashed line sets the insurance use of offers to zero. Outcomes are deviations from the no-shock path.

Figure 3. Impulse Response to a Perceived Job-Loss Risk Shock

The two lines in Figure 3 show the role of offer usability. When offers can insure separations, worse beliefs raise search and slightly reduce E-to-U transitions. When offers cannot be used at separation, search still rises, but unemployment does not fall. Beliefs move search through the value of an outside offer. Unemployment moves only if that offer can keep a separated worker employed.

Appendix Table A26 reports a related unemployment-insurance counterfactual. Raising recession-state benefits crowds out precautionary employed search, and the decline in covered E-to-E moves is largest when a larger share of offers can insure job loss. However, the unemployment response is driven mostly by the standard UI surplus, wage, and tightness channel. The search-crowd-out component raises recession unemployment by only 0.005–0.029 percentage points, so I treat the UI exercise as a supporting policy implication rather than part of the core quantitative arc.

5.3 Discussion

The counterfactuals put the empirical results in perspective. The survey evidence shows a strong behavioral margin: workers who perceive higher job-loss risk search more while still employed. The model helps disentangle the mechanism behind this relationship. Beliefs affect search through a precautionary motive—when workers think job loss is more likely, an outside offer becomes more valuable because it can keep a separation from becoming an unemployment spell.

This mechanism also helps explain countercyclical search. In recessions, lower tightness makes offers harder to get, which by itself would discourage search. At the same time, perceived job-loss risk is higher, so the insurance value of an offer rises. When this precautionary force is strong enough, employed search rises even though the market is less favorable.

The same mechanism does not imply a large aggregate unemployment effect by itself. Search affects unemployment only through offers that are available at the moment of separation. If an extra offer would have produced an ordinary job-to-job move anyway, it changes the employer a worker moves to but does not prevent an unemployment spell. If the offer is available when a job is lost, it can turn a potential E-to-U transition into an E-to-E transition. The unemployment effect depends on the composition of outside offers, not only on how much search rises.

In the baseline model, most offers serve the insurance role by construction, so precautionary search offsets a meaningful share of the recessionary rise in unemployment. Once the model also matches ordinary E-to-E mobility, many offers are instead assigned to ladder mobility. Beliefs still move search, and the model generates countercyclical employed search, but the aggregate effect becomes smaller because only part of the additional search is useful as insurance against job loss.

The main takeaway is that macro and personal beliefs matter for workers' choices, and they help explain why employed search rises when the economy weakens. Beliefs matter for aggregate unemployment only to the extent that the search they induce produces offers that can be used at separation. The model makes that distinction explicit by separating search for better jobs from search that insures job loss.

6 Conclusion

This paper studies how beliefs about the economy affect job search while workers are still employed. The empirical results point to a simple channel. Workers who are more pessimistic about the macroeconomy perceive a higher risk of losing their own job, and workers who perceive higher job-loss risk search more. In the Survey of Consumer Expectations, perceived job-loss risk predicts search intensity, forecasts subsequent separations, and predicts job-security motives among workers who search. The response is also strongest when workers see both high job-loss risk and weak re-employment prospects, exactly the case in which having an outside offer is most valuable. The ECB Consumer Expectations Survey shows the same first-order pattern in a different setting: aggregate pessimism predicts perceived job-loss risk, and perceived job-loss risk predicts search.

I interpret these facts as evidence of precautionary on-the-job search. Employed workers search not only to move up the job ladder, but also to reduce the chance that job loss turns into unemployment. The model formalizes this distinction. Actual job-loss risk governs realized flows into unemployment, while perceived job-loss risk governs the worker's search decision. When workers think job loss is more likely, an outside offer becomes more valuable as insurance, so they search more.

The model also clarifies the aggregate margin. Beliefs about job-loss risk help explain why employed search rises in recessions despite lower tightness and worse offer prospects. But the effect on unemployment depends on what the extra search produces. Search lowers unemployment only when the additional offer is available at the moment of separation; otherwise, it changes job-to-job mobility without changing unemployment.

The main contribution is to separate two margins that are often bundled together. Aggregate beliefs matter for employed workers' search decisions, and they matter through perceived job-loss risk. Whether they also have a large effect on unemployment depends on how much of that search actually insures workers against job loss.

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A Data Appendix: Survey of Consumer Expectations (SCE)

This appendix provides additional detail on the Survey of Consumer Expectations (SCE) and the construction of the empirical analysis sample. The main text documents that perceived job-loss risk predicts on-the-job search intensity and motivates a precautionary interpretation. Since that interpretation depends on how expectations and search are measured, I describe the survey design, variable construction, and sample selection here.

A.1 Survey design and panel structure

The SCE is a nationally representative monthly internet panel administered by the Federal Reserve Bank of New York since 2013. Approximately 1,300 household heads participate each month. Respondents remain in the panel for up to 12 consecutive months, allowing expectations and realized outcomes to be linked within person. All reported statistics use SCE survey weights. Regressions cluster standard errors at the individual level to account for within-person serial correlation.

The panel structure matters for two reasons emphasized in the main text. First, it allows validation of beliefs against subsequent realizations (Section 2.4). Second, it permits within-person specifications, reported below, that address concerns about stable individual heterogeneity jointly driving expectations and search.

A.2 Core survey and labor-market expectations

The monthly *core survey* elicits probabilistic expectations on a 0–100 scale. The analysis uses four measures:

1. *Macroeconomic pessimism:* the subjective probability that the U.S. unemployment rate will be higher in 12 months.

2. *Job-loss probability:* the subjective probability of involuntarily losing the current job within 12 months.

3. *Voluntary separation probability:* the subjective probability of leaving the job voluntarily within 12 months.

4. *Job-finding probability conditional on job loss:* the subjective probability of finding a job within three months if the current job is lost.

All expectations are rescaled to $[0, 1]$. The distinction between involuntary job-loss beliefs and voluntary separation beliefs is economically important. The model's precautionary motive operates through perceived *inflow risk* into unemployment. The empirical analysis therefore distinguishes involuntary risk from voluntary mobility and from beliefs about re-employment prospects.

A.3 Labor Market Module and search measurement

The *Labor Market Module* (LMM) is administered every four months to respondents while in the panel. The primary outcome variable is weekly job-search intensity: total hours spent searching or applying for jobs in the past seven days. For each LMM interview month, I match expectations from the core survey in the same calendar month (or the most recent prior core-survey response if the same-month observation is missing).

Search hours are winsorized at the 5th and 95th percentiles to limit the influence of extreme reports. The right-skewed distribution motivates robustness checks using alternative functional forms (Appendix Table A9). The model interprets search as an intensity choice, not literal hours; accordingly, the calibration targets the *semi-elasticity* of search with respect to perceived job-loss risk rather than the level of hours.

The LMM also records industry of current employment (`LMind`). This variable is used to construct industry fixed effects and industry×time fixed effects in robustness specifications reported below. These specifications address the concern that unobserved industry-level shocks may jointly affect beliefs and search.

A.4 Appendix Table A1: Summary Statistics

Appendix Table A1 reports weighted summary statistics for the employed LMM sample.

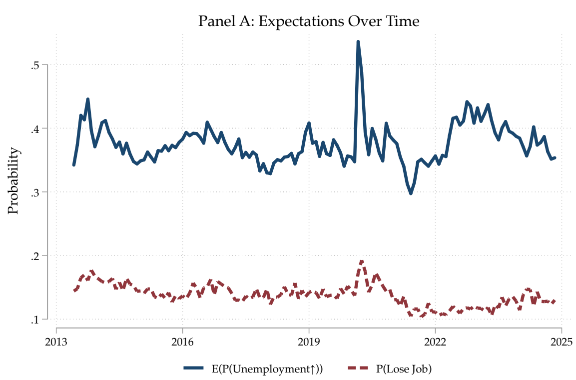
Two features are worth emphasizing. First, subjective expectations display substantial dispersion. The cross-sectional standard deviation of job-loss beliefs is comparable to its mean. This dispersion underlies the main empirical design: within a given state and month, workers hold different beliefs about separation risk.

Second, search intensity is highly skewed. The mean is approximately 3.8 hours per week, but the standard deviation exceeds the mean. This motivates both winsorization and the use of alternative specifications in Appendix Table A9. The empirical results in the main text do not depend on functional form.

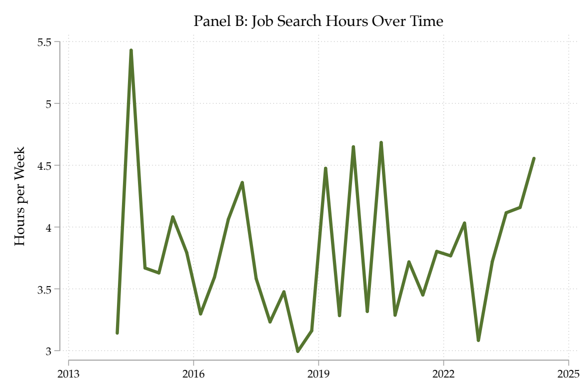
Table A1. Summary Statistics: Employed Workers

	N	Mean	SD
Search hours (winsorized)	4226	3.774	4.800
$E_{t-1}(\Delta\,\Pr(U_{t+12} \geq 0))$	4217	0.394	0.239
$E_{t-1}(\Pr(\text{Lose Job}_{t+12}))$	4220	0.213	0.234
$E_{t-1}(\Pr(\text{Leave Job}_{t+12}))$	4220	0.402	0.304
$E_{t-1}(\Pr(\text{Find}\vert\text{Lose}_{t+3}))$	4219	0.571	0.300
$E_{t-1}(\pi_{t+12})$ (point)	4219	0.072	0.213
Age	4226	41.508	11.911
Female	4224	0.453	0.498
Married	4225	0.599	0.490
White	4226	0.772	0.419
Black	4226	0.145	0.352
Hispanic	4225	0.120	0.325
HH income: Under 50k	4211	0.376	0.484
HH income: 50k–100k	4211	0.324	0.468
HH income: Over 100k	4211	0.300	0.459
Education: High school	4205	0.240	0.427
Education: Some college	4205	0.293	0.455
Education: College+	4205	0.467	0.499

Sample: Employed workers with non-missing search hours (SCE Labor Market Module).
Search hours winsorized at 5th and 95th percentiles.
Means/SD computed with SCE weights (analytic weights); N is unweighted non-missing count.
Binary indicators report sample shares (means).



Expectations Over Time



Search Hours

Notes: The figure plots weighted monthly means of macro unemployment expectations (core survey) and weekly search hours (LMM) for employed respondents. Expectations spike during major aggregate events (notably COVID-19). Search hours are more volatile because the LMM is only administered every four months to a rotating subsample. The aggregate time series is therefore noisy, motivating the use of within-state-month cross-sectional variation in the main analysis.

Figure A1. Time Series of Expectations and Search Intensity

A.5 Appendix Figure A1: Aggregate Time Series

B Empirical Appendix

This section collects robustness and identification checks referenced in Section 2. Each exhibit addresses a specific concern about the interpretation of the belief–search relationship. The question throughout is whether the positive association between perceived job-loss risk and on-the-job search intensity reflects a precautionary response to perceived separation risk, or instead captures omitted local shocks, sectoral conditions, functional-form artifacts, or stable individual heterogeneity.

B.1 Alternative Fixed Effects

The specification replaces separate state and time fixed effects with their full interaction:

$$\text{SearchHours}_{ist} = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it}\gamma + \mu_{st} + \varepsilon_{ist}, \tag{12}$$

where $\mathbf{E}_{it}$ collects the expectation variables (added progressively across columns), $\mathbf{X}_{it}$ are demographic controls, and μ_{st} denotes state×time fixed effects. Standard errors are clustered at the individual level; observations are weighted by survey weights. The sample consists of employed workers.

State×time fixed effects. This specification absorbs all state-level labor-market conditions in a given month, including unemployment rates, vacancy conditions, and other local shocks. Identification comes from cross-sectional differences in beliefs among workers residing in the *same state in the same month*.

The stability of the job-loss coefficient relative to the baseline indicates that the belief–search relationship is not driven by common local shocks. Even within a fixed local environment, workers who perceive higher separation risk search more intensively.

This specification replaces time fixed effects with industry×time interactions and retains state fixed effects:

$$\text{SearchHours}_{ijst} = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it}\gamma + \mu_{s} + \delta_{jt} + \varepsilon_{ijst},$$

where δ_{jt} denotes industry×time fixed effects (using the LMM industry classification) and μ_s are state fixed effects.

Table A2. Expectations and On-the-Job Search Intensity

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
$E_t[\Delta \Pr(U_{t+12} \geq 0)]$	1.538***	0.920**	0.906**	0.814*	0.870**
	(0.420)	(0.415)	(0.412)	(0.435)	(0.408)
$E_t \Pr(\text{Lose Job}_{t+12})$		2.064***	1.841***	1.533***	2.719***
		(0.499)	(0.502)	(0.517)	(0.991)
$E_t \Pr(\text{Leave Job}_{t+12})$		1.372***	1.691***	1.040**	1.692***
		(0.398)	(0.407)	(0.439)	(0.407)
$E_t \Pr(\text{Find Job}_{t+3})$			-1.259***	-1.762***	-0.973**
			(0.352)	(0.360)	(0.441)
$E_t \Pr(\text{Offer}_{t+3})$				0.020***	
				(0.004)	
$E_t(\Pr(\text{Lose Job}_{t+12}))\times$ $E_t(\Pr(\text{Find Job}_{t+3}))$					-1.523
					(1.452)
$E_t(\pi_{t+12})$	-0.019	-0.019	-0.023	-0.019	-0.024
	(0.023)	(0.023)	(0.023)	(0.024)	(0.023)
Separation expectations	No	Yes	Yes	Yes	Yes
Job-finding expectations	No	No	P(find) given lose	Both	Interaction
State × Time FE	Yes	Yes	Yes	Yes	Yes
Observations	4132	4131	4130	3922	4130
R-squared	0.391	0.405	0.409	0.419	0.409

Notes: Dependent variable: weekly search hours (winsorized). Expectations scaled to $[0,1]$. Specification absorbs state×time fixed effects; controls include age polynomial, education, race, gender, marital status, household income; weighted by survey weights; standard errors clustered by individual; sample: employed workers. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Industry×time fixed effects. Industry×time fixed effects absorb sector-specific shocks at a monthly frequency, addressing the concern that industry downturns jointly raise perceived risk and search.

The job-loss coefficient remains economically and statistically meaningful under this demanding specification. The belief–search relationship is not simply reflecting industry-wide conditions; it also reflects heterogeneity in perceived risk among workers in the same industry at the same point in time.

The sample is restricted to employed workers with non-missing industry. Column (1) uses state×time fixed effects (μ_{st}); Column (2) adds industry fixed effects (δ_j); Column (3) replaces these with industry×time fixed effects (δ_{jt}) alongside state fixed effects (μ_s).

Table A3. Expectations and On-the-Job Search Intensity

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
E_t $\Delta\,\Pr(U_{t+12} \geq 0)$	1.671***	0.981**	1.010**	1.110**	1.006**
	(0.452)	(0.435)	(0.435)	(0.454)	(0.432)
E_t $\Pr(\text{Lose Job}_{t+12})$		2.107***	1.844***	1.652***	1.976**
		(0.483)	(0.493)	(0.515)	(0.975)
E_t $\Pr(\text{Leave Job}_{t+12})$		1.595***	1.917***	1.411***	1.917***
		(0.369)	(0.379)	(0.415)	(0.379)
E_t $\Pr(\text{Find Job}_{t+3})$			-1.222***	-1.566***	-1.176***
			(0.342)	(0.358)	(0.441)
E_t $\Pr(\text{Offer}_{t+3})$				0.014***	
				(0.004)	
$E_t\,\Pr(\text{Lose Job}_{t+12})$ $\times E_t\,\Pr(\text{Find Job}_{t+3})$					-0.227
					(1.432)
$E_t(\pi_{t+12})$ (dist. mean)	-0.014	-0.019	-0.023	-0.020	-0.023
	(0.027)	(0.027)	(0.027)	(0.029)	(0.027)
Separation expectations	No	Yes		Yes	Yes
Job-finding expectations	No	No	P(find) given lose	Both	Interaction
Industry × Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	3086	3085	3085	2925	3085
R-squared	0.332	0.350	0.354	0.361	0.354

Notes: Dependent variable: weekly search hours (winsorized). Expectations scaled to $[0,1]$. Specification absorbs industry×time fixed effects and includes state FE; controls include age polynomial, education, race, gender, marital status, household income; weighted by survey weights; standard errors clustered by individual; sample: employed workers. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A4. Main Result Robustness: Industry Controls (LMind)

	(1) State×Time FE	(2) + Industry FE	(3) Industry×Time FE
$E_t[\Delta \Pr(U_{t+12} \geq 0)]$	0.853*	0.772	1.006**
	(0.499)	(0.508)	(0.432)
$E_t \Pr(\text{Lose Job}_{t+12})$	1.310	1.379	1.976**
	(1.152)	(1.161)	(0.975)
$E_t \Pr(\text{Leave Job}_{t+12})$	2.104***	2.068***	1.917***
	(0.464)	(0.464)	(0.379)
$E_t \Pr(\text{Find Job}_{t+3})$	-1.433**	-1.356**	-1.176***
	(0.588)	(0.575)	(0.441)
$E_t \Pr(\text{Lose Job}_{t+12}) \times$	0.466	0.460	-0.227
$E_t \Pr(\text{Find Job}_{t+3})$	(1.775)	(1.769)	(1.432)
$E_t(\pi_{t+12})$	-0.017	-0.014	-0.023
	(0.029)	(0.029)	(0.027)
State × Time FE	Yes	Yes	No
Industry × Time FE	No	No	Yes
Industry FE	No	Yes	No
Observations	3085	3085	3085
R-squared	0.489	0.495	0.354

Notes: Sample restricted to employed workers with non-missing LMind industry. Column 1 absorbs state×time FE; column 2 adds industry FE; column 3 absorbs industry×time FE and includes state FE. Expectations scaled to $[0, 1]$; weighted by survey weights; standard errors clustered by individual. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Industry and state controls combined. Adding industry fixed effects alongside time and state controls further isolates within-industry, within-state belief variation. The magnitude and sign of the key coefficients are preserved. These specifications show that the precautionary relationship survives controls for both geographic and sectoral conditions.

These exercises do not establish causality. They show that the empirical relationship is not explained by observable local or sectoral shocks, strengthening the case that belief heterogeneity plays an independent role.

B.2 COVID Exclusion and Two-Way Clustered Standard Errors

Because the sample spans the COVID-19 recession, Appendix Tables below repeat the main SCE search regressions excluding April–December 2020. I also report versions with standard errors two-way clustered by individual and survey month, estimated with explicit state and month fixed effects. The coefficients on perceived job-loss risk remain positive, economically large, and statistically precise.

Table A5. Expectations and On-the-Job Search Intensity: Robustness

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
$E\_t(\Delta \Pr(U\_t + 12 \geq 0))$	1.581***	0.898**	0.900**	0.819**	0.869**
	(0.379)	(0.363)	(0.362)	(0.377)	(0.359)
E_t(Pr(Lose Job_t + 12))		2.083***	1.854***	1.665***	2.871***
		(0.432)	(0.436)	(0.449)	(0.845)
E_t(Pr(Leave Job_t + 12))		1.587***	1.922***	1.342***	1.930***
		(0.339)	(0.343)	(0.372)	(0.343)
E_t(Pr(Find|Lose_t + 3))			-1.210***	-1.628***	-0.878**
			(0.300)	(0.304)	(0.370)
E_t(Pr(Offer_t + 3))				0.019***	
				(0.003)	
E_t(Pr(Lose_t + 3)) × E_t(Pr(Find|Lose_t + 3))					-1.765
					(1.236)
$E\_t(\pi\_t + 12)$ (dist. mean)	-0.023	-0.023	-0.026	-0.022	-0.027
	(0.022)	(0.022)	(0.023)	(0.024)	(0.023)
Sample	No COVID	No COVID	No COVID	No COVID	No COVID
Separation expectations	No	Yes	Yes	Yes	Yes
Job-finding expectations	No	No	P(find|lose)	Both	Interaction
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	3968	3967	3966	3758	3966
R-squared	0.123	0.147	0.152	0.162	0.153

Standard errors in parentheses
Dependent variable: Weekly job search hours (winsorized).
Sample excludes April–December 2020. Standard errors clustered by individual.
Controls: Age polynomial, education, race, gender, marital status, household income.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Current-wage control. Appendix Table A8 adds log current wage to the main SCE search regressions, using the common sample of employed workers with positive reported current wages. This check asks whether the baseline relationship is simply picking up the fact that low-wage workers search more. The answer is no. The coefficients on macro pessimism and perceived job-loss risk are essentially unchanged after controlling for log current wage, while log current wage itself is negatively associated with search hours.

B.3 Alternative Outcomes and Panel Checks

Columns (1)–(3) estimate the baseline specification with alternative dependent variables:

$$f(\text{SearchHours}_{ist}) = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it}\gamma + \mu_s + \lambda_t + \varepsilon_{ist},$$

Table A6. Expectations and On-the-Job Search Intensity: Robustness

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
$E\_t(\Delta\,\mathrm{Pr}(U\_t+12 \geq 0))$	1.607***	0.953***	0.953***	0.872**	0.928***
	(0.362)	(0.353)	(0.355)	(0.352)	(0.353)
$E\_t(\mathrm{Pr}(\text{Lose Job}\_t+12))$		2.046***	1.836***	1.629***	2.682***
		(0.433)	(0.431)	(0.449)	(0.821)
$E\_t(\mathrm{Pr}(\text{Leave Job}\_t+12))$		1.567***	1.884***	1.313***	1.890***
		(0.355)	(0.368)	(0.355)	(0.372)
$E\_t(\mathrm{Pr}(\text{Find}\mid\text{Lose}\_t+3))$			-1.156***	-1.568***	-0.877***
			(0.261)	(0.281)	(0.319)
$E\_t(\mathrm{Pr}(\text{Offer}\_t+3))$				0.019***	
				(0.003)	
$E\_t(\mathrm{Pr}(\text{Lose}\_t+3)) \times E\_t(\mathrm{Pr}(\text{Find}\mid\text{Lose}\_t+3))$					-1.479
					(1.097)
$E\_t(\pi\_t+12)$ (dist. mean)	-0.025	-0.025	-0.028	-0.024	-0.029
	(0.019)	(0.019)	(0.019)	(0.021)	(0.019)
Sample	Full	Full	Full	Full	Full
Separation expectations	No	Yes	Yes	Yes	Yes
Job-finding expectations	No	No	P(find|lose)	Both	Interaction
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	4132	4131	4130	3922	4130
R-squared	0.125	0.148	0.152	0.163	0.153

Standard errors in parentheses

Dependent variable: Weekly job search hours (winsorized).

Sample: Employed workers. Standard errors two-way clustered by individual and survey month.

Controls: Age polynomial, education, race, gender, marital status, household income.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

where $f(\cdot)$ is the identity (levels), $\log(1+\cdot)$, or the arc-percent transformation $(y-\bar{y})/(0.5(y+\bar{y}))$, with $\bar{y}$ denoting mean search hours in the estimation sample. Column (4) estimates a Poisson quasi-maximum likelihood model with the same regressors and fixed effects.

Functional form robustness. Search hours are highly skewed, with a substantial mass at zero. Appendix Table A9 shows that the positive association between perceived job-loss probability and search intensity is robust across alternative transformations and specifications, including $\log(1+y)$, the arc-percent transformation, and Poisson quasi-maximum likelihood.

The dependent variable is an indicator for any positive search hours:

$$\mathbb{1}\{\text{SearchHours}_{ist} > 0\} = \beta'\mathbf{E}_{it} + \mathbf{X}_{it}'\gamma + \mu_s + \lambda_t + \varepsilon_{ist}.$$

Table A7. Expectations and On-the-Job Search Intensity: Robustness

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
$E\_t(\Delta \Pr(U\_t + 12 \geq 0))$	1.581***	0.898**	0.900**	0.819**	0.869**
	(0.373)	(0.363)	(0.365)	(0.366)	(0.363)
E_t(Pr(Lose Job_t + 12))		2.083***	1.854***	1.665***	2.871***
		(0.441)	(0.435)	(0.451)	(0.781)
E_t(Pr(Leave Job_t + 12))		1.587***	1.922***	1.342***	1.930***
		(0.371)	(0.385)	(0.366)	(0.390)
E_t(Pr(Find|Lose_t + 3))			-1.210***	-1.628***	-0.878***
			(0.269)	(0.291)	(0.309)
E_t(Pr(Offer_t + 3))				0.019***	
				(0.003)	
E_t(Pr(Lose_t + 3)) × E_t(Pr(Find|Lose_t + 3))					-1.765*
					(1.060)
$E\_t(\pi\_t + 12)$ (dist. mean)	-0.023	-0.023	-0.026	-0.022	-0.027
	(0.020)	(0.020)	(0.020)	(0.022)	(0.020)
Sample	No COVID	No COVID	No COVID	No COVID	No COVID
Separation expectations	No	Yes	Yes	Yes	Yes
Job-finding expectations	No	No	P(find|lose)	Both	Interaction
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	3968	3967	3966	3758	3966
R-squared	0.123	0.147	0.152	0.162	0.153

Standard errors in parentheses
Dependent variable: Weekly job search hours (winsorized).
Sample excludes April–December 2020. Standard errors two-way clustered by individual and survey month.
Controls: Age polynomial, education, race, gender, marital status, household income.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Extensive margin. Appendix Table A10 shows that perceived job-loss probability significantly predicts the probability of engaging in any search at all. Macro pessimism loses statistical significance once personal beliefs are included, consistent with the separation-risk-first interpretation emphasized in the main text.

Two within-person specifications are estimated. First, a first-difference specification:

$$\Delta \text{SearchHours}_{it} = \beta' \Delta \mathbf{E}_{it} + \lambda_t + \Delta \varepsilon_{it}.$$

Second, an individual fixed-effects specification:

$$\text{SearchHours}_{it} = \beta' \mathbf{E}_{it} + \alpha_i + \lambda_t + \varepsilon_{it},$$

where α_i denotes individual fixed effects. Standard errors are clustered at the individual level.

Table A8. Main Search Regressions with Log Current Wage Control

	Main	+ ln wage	Offer	+ ln wage	Interaction	+ ln wage
$E_t(\Delta\,\Pr(U_{t+12} \geq 0))$	0.880**	0.912***	0.780**	0.824**	0.854**	0.886**
	(0.349)	(0.349)	(0.362)	(0.363)	(0.345)	(0.345)
$E_t(\Pr(\text{Lose Job}_{t+12}))$	1.804***	1.797***	1.594***	1.600***	2.679***	2.670***
	(0.432)	(0.429)	(0.444)	(0.440)	(0.839)	(0.833)
$E_t(\Pr(\text{Leave Job}_{t+12}))$	1.887***	1.874***	1.281***	1.290***	1.893***	1.880***
	(0.333)	(0.334)	(0.362)	(0.361)	(0.333)	(0.334)
$E_t(\Pr(\text{Find}\|\text{Lose}_{t+3}))$	-1.161***	-1.132***	-1.579***	-1.545***	-0.871**	-0.842**
	(0.297)	(0.296)	(0.300)	(0.298)	(0.369)	(0.369)
$E_t(\Pr(\text{Offer}_{t+3}))$			0.020***	0.019***		
			(0.003)	(0.003)		
$E_t(\Pr(\text{Lose}_{t+3})) \times$ $E_t(\Pr(\text{Find}\|\text{Lose}_{t+3}))$					-1.529	-1.524
					(1.229)	(1.225)
Log current wage		-0.235**		-0.257**		-0.235**
		(0.112)		(0.119)		(0.112)
$E_t(\pi_{t+12})$ (dist. mean)	-0.029	-0.030	-0.024	-0.024	-0.030	-0.031
	(0.022)	(0.022)	(0.023)	(0.023)	(0.022)	(0.022)
Log wage control	No	Yes	No	Yes	No	Yes
Observations	4088	4088	3882	3882	4088	4088
R-squared	0.152	0.156	0.163	0.168	0.152	0.156

Notes: The dependent variable is weekly job search hours, winsorized. The sample is restricted to employed workers with non-missing positive current wage. All specifications include state and time fixed effects and the baseline demographic controls, including household income. Standard errors, clustered by individual, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Within-person variation. Individual fixed-effects and first-difference specifications exploit within-person variation in beliefs over time. Statistical power is limited because most respondents appear only once or twice in the LMM. Consistent with this limitation, the within-person estimates are imprecise. They are nonetheless broadly consistent in sign with the cross-sectional evidence, particularly for voluntary separation beliefs.

B.4 Employed vs. Unemployed and Heterogeneity

The baseline specification is estimated separately for employed and unemployed workers. For the unemployed, the expectation vector includes macro pessimism and perceived job-finding probability (job-loss beliefs are not applicable):

$$\text{SearchHours}_{ist} = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it}\gamma + \mu_s + \lambda_t + \varepsilon_{ist}.$$

Employment status comparison. Appendix Table A12 compares belief–search relationships across employment status. The belief–search relationship is strongest among employed workers, consistent with the interpretation that on-the-job search serves an

Table A9. Robustness: Alternative Dependent Variable Specifications

	(1) Levels	(2) Log	(3) Arc-%	(4) Poisson
main				
$E\_t(\Delta\,\mathrm{Pr}(U\_t + 12 \geq 0))$	0.953***	0.130**	0.144**	0.246***
	(0.355)	(0.058)	(0.070)	(0.091)
$E\_t(\mathrm{Pr}(\text{Lose Job}\_t + 12))$	1.836***	0.344***	0.413***	0.398***
	(0.429)	(0.065)	(0.077)	(0.094)
$E\_t(\mathrm{Pr}(\text{Leave Job}\_t + 12))$	1.884***	0.389***	0.479***	0.512***
	(0.333)	(0.052)	(0.062)	(0.083)
$E\_t(\mathrm{Pr}(\text{Find}\vert\text{Lose}\_t + 3))$	-1.156***	-0.210***	-0.251***	-0.351***
	(0.294)	(0.050)	(0.060)	(0.078)
Dep. Variable	Levels	Log(1+y)	Arc-%	Poisson
Observations	4130	4130	4130	4130
R-squared	0.152	0.159	0.156	

Notes: Dependent variable variants: levels, log(1+y), arc-% $(y - \bar{y})/(0.5(y + \bar{y}))$, and Poisson. Expectations scaled to $[0, 1]$. All specifications include state and time fixed effects plus controls; weighted by survey weights; standard errors clustered by individual (Poisson uses robust SE). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

insurance role against separation risk.

The baseline specification is estimated on subsamples defined by marital status, median income split, and age (< 40 vs. ≥ 40).

Worker heterogeneity. Appendix Table A13 explores heterogeneity by marital status, income, and age. The belief–search relationship is present across subgroups and is not confined to a narrow segment of the labor market.

B.5 Additional Validation of the Separation-Risk Channel

To further assess whether subjective job-loss beliefs contain information about realized labor-market risk, this subsection reports two additional validation exercises that are not used for identification in the main search regressions.

Prediction at multiple horizons. Appendix Table A14 re-estimates the validation specification at horizons from 1 to 9 months and with two right-censored outcomes for "any non-employment within 12 months." The coefficient on P(Lose Job) is positive and precisely estimated at short and medium horizons, and remains positive in the censored

Table A10. Expectations and On-the-Job Search: Extensive Margin

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
E_t $\Delta \Pr(U_{t+12} \geq 0)$	0.039	0.014	0.015	0.003	0.013
	(0.026)	(0.026)	(0.026)	(0.026)	(0.026)
E_t $\Pr(\text{Lose Job}_{t+12})$		0.072^{***}	0.066^{***}	0.057^{**}	0.101^{**}
		(0.024)	(0.024)	(0.025)	(0.051)
E_t $\Pr(\text{Leave Job}_{t+12})$		0.064^{***}	0.073^{***}	0.040	0.073^{***}
		(0.021)	(0.022)	(0.024)	(0.022)
E_t $\Pr(\text{Find Job}_{t+3})$			-0.033	-0.054^{**}	-0.022
			(0.023)	(0.023)	(0.029)
E_t $\Pr(\text{Offer}_{t+3})$				0.001^{***}	
				(0.000)	
$E_t \Pr(\text{Lose Job}_{t+12})$ $\times E_t \Pr(\text{Find Job}_{t+3})$					-0.062
					(0.081)
$E_t(\pi_{t+12})$ (dist. mean)	-0.002	-0.002	-0.002	-0.002	-0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Separation expectations	No	Yes	Yes	Yes	Yes
Job-finding expectations	No	No	P(find) given lose	Both	Interaction
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	4132	4131	4130	3922	4130
R-squared	0.045	0.051	0.052	0.061	0.052

Notes: Dependent variable: indicator for any search (> 0 hours). Expectations scaled to $[0, 1]$. State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual; sample: employed workers. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A11. Within-Person Identification: First Differences and Fixed Effects

	(1) FD Base	(2) FD + Sep.	(3) FE Base	(4) FE + Sep.
$\Delta E_t[\Delta \Pr(U_{t+12} \geq 0)]$	1.335	0.905		
	(0.957)	(0.918)		
Lagged $E_t[\Delta \Pr(U_{t+12} \geq 0)]$			0.404	-0.201
			(0.543)	(0.558)
ΔE_t Pr(Lose Job_{t+12})		1.334		
		(1.034)		
Lagged E_t Pr(Lose Job_{t+12})				1.103
				(0.834)
ΔE_t Pr(Leave Job_{t+12})		2.450^{***}		
		(0.831)		
Lagged E_t Pr(Leave Job_{t+12})				1.875^{***}
				(0.514)
Specification	First Diff	First Diff	Individual FE	Individual FE
Observations	1183	1111	4576	4402
R-squared	0.069	0.079	0.029	0.045

Notes: Columns 1–2 use first differences; columns 3–4 include individual fixed effects. Expectations scaled to $[0, 1]$. Time controls as in main specs; standard errors clustered by individual; short panel (average 1.4 LMM observations per worker) limits within-person power. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

12-month outcomes. The belief measure tracks realized separation-related risk over horizons that are relevant for employed search decisions.

Outcome-definition robustness. Appendix Table A15 compares two t+3 outcomes: broad non-employment/attachment and strict unemployment. The job-loss belief coefficient is positive and statistically significant in both columns. The precautionary interpretation does not depend on one narrow coding of post-separation labor-market status.

Direct re-employment after displacement. Appendix Table A16 uses the CPS Displaced Worker Supplement from IPUMS CPS to examine the conversion margin among displaced workers (Flood et al., 2025). Following the DWS displacement literature (e.g., Farber, 2017), the sample includes workers ages 20–64 displaced for plant closing or move, insufficient work, or position or shift abolished, who are re-employed full time at the interview. The outcome equals one if the worker reports zero weeks without work between the lost job and the current job; the robustness outcome equals one for four or fewer weeks. Recession

Table A12. Expectations and Search: Employed vs. Unemployed Workers

	(1) Employed	(2) Unemployed
$E\_t(\Delta\,\mathrm{Pr}(U\_t + 12 \geq 0))$	0.953***	0.181
	(0.355)	(1.318)
$E\_t(\mathrm{Pr}(\text{Lose Job}\_t + 12))$	1.836***	
	(0.429)	
$E\_t(\mathrm{Pr}(\text{Leave Job}\_t + 12))$	1.884***	
	(0.333)	
$E\_t(\mathrm{Pr}(\text{Find}\vert\text{Lose}\_t + 3))$	-1.156***	
	(0.294)	
$E\_t(\mathrm{Pr}(\text{Find}\vert\text{Unemp}\_t + 3))$		2.130**
		(1.018)
$E\_t(\pi\_t + 12)$ (dist. mean)	-0.028	0.134***
	(0.022)	(0.039)
Sample	Employed	Unemployed
Observations	4130	712
R-squared	0.152	0.295

Notes: Employed sample: search hours on separation and job-finding expectations; unemployed sample: search hours on macro expectations and job-finding probability. Expectations scaled to $[0, 1]$. State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

displacement years and increases in national unemployment both predict lower direct re-employment, even after controlling for displacement reason, years since displacement, and lost-job occupation and industry fixed effects. Post-1994 estimates are similar, so the DWS recall-window change is not driving the result.

B.6 ECB Consumer Expectations Survey Validation

Appendix Table A17 reports external-validation exercises in the ECB Consumer Expectations Survey, using the April 2026 release through 2026Q1. Panels A and B use employed respondents in the regular quarterly labor-market module matched to monthly expectations. Panel A asks whether aggregate labor-market pessimism predicts perceived job-loss risk. Panel B asks whether perceived job-loss risk predicts search outcomes. Panel C reports a supplementary heterogeneity check. The regular quarterly module does not

Table A13. Heterogeneity by Marital Status, Income, and Age

	(1) Unmarried	(2) Married	(3) Low Inc.	(4) High Inc.	(5) Young	(6) Old
$E_t[\Delta \Pr(U_{t+12} \geq 0)]$	1.125^{*}	0.887^{**}	1.254^{***}	-0.030	1.272^{**}	0.647
	(0.658)	(0.408)	(0.467)	(0.464)	(0.516)	(0.483)
$E_t \Pr(\text{Lose Job}_{t+12})$	2.525^{***}	0.843	1.416^{**}	2.243^{***}	1.587^{**}	1.906^{***}
	(0.672)	(0.552)	(0.572)	(0.628)	(0.696)	(0.578)
$E_t \Pr(\text{Leave Job}_{t+12})$	1.698^{***}	2.311^{***}	2.100^{***}	1.723^{***}	2.090^{***}	1.985^{***}
	(0.557)	(0.419)	(0.462)	(0.442)	(0.502)	(0.473)
$E_t \Pr(\text{Find Job}_{t+3})$	-1.173^{***}	-1.298^{***}	-1.366^{***}	-0.510	-0.862^{*}	-1.449^{***}
	(0.455)	(0.402)	(0.400)	(0.401)	(0.466)	(0.374)
Sample	Unmarried	Married	Low Income	High Income	Age < 40	Age >= 40
Observations	1675	2512	2844	1343	1989	2198
R-squared	0.221	0.189	0.164	0.139	0.187	0.207

Notes: Subsamples by marital status, income (median split), and age (< 40 vs ≥ 40). Expectations scaled to $[0, 1]$. State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A14. Validation: Beliefs Predict Separations at Multiple Horizons (Q10-based)

	(1) 1m	(2) 3m	(3) 6m	(4) 9m	(5) 12m exact	(6) Any<=12 (fup>=1)	(7) Any<=12 (fup>=6)
P(Lose Job)	0.0770^{***}	0.1290^{***}	0.1487^{***}	0.1266^{***}	0.5101	0.1996^{***}	0.2504^{***}
	(0.0065)	(0.0107)	(0.0156)	(0.0198)	(0.3434)	(0.0159)	(0.0245)
Outcome	Not employed/attached	Not employed/attached	Not employed/attached	Not employed/attached	Not employed/attached (exact; selected)	Any not-employed (censored), fup≥1	Any not-employed (censored), fup≥6
Horizon	1 month	3 months	6 months	9 months	12 months	≤12 months	≤12 months
Observations	71116	55832	34656	16586	290	71116	27032
R-squared	0.171	0.205	0.287	0.415	0.690	0.206	0.349

Exact-horizon outcomes require observation at t+h.

Any<=12 is right-censored and uses observed forward months; sample restrictions based on consecutive follow-up fup_consec12.

Table A15. Robustness: Beliefs Predict Both Non-Employment and Unemployment (Q10-based)

	(1) Not employed/attached (t+3)	(2) Unemployed (t+3)
P(Lose Job)	0.1290^{***}	0.0726^{***}
	(0.0107)	(0.0079)
Observations	55832	55832
R-squared	0.205	0.186

Table A16. DWS Conversion Margin: Weeks Without Work After Displacement

	(1) Zero weeks	(2) Zero weeks	(3) Zero weeks	(4) ≤ 4 weeks	(5) ≤ 4 weeks	(6) ≤ 4 weeks
Recession year	-0.034***			-0.083***		
	(0.010)			(0.027)		
Δ national unemployment		-0.012***			-0.032***	
		(0.003)			(0.005)	
State unemployment			0.005			0.004
			(0.004)			(0.003)
Sample	National	National: dU	State	National	National: dU	State
Observations	22713	22428	19821	22713	22428	19821
R-squared	0.069	0.071	0.083	0.091	0.095	0.108

Standard errors in parentheses

Sample: ages 20–64, displaced for plant closing/move, insufficient work, or position/shift abolished, and reemployed full-time at interview.

Displacement year is survey year minus DWLASTWRK.

National regressions include demographics, reason-for-displacement indicators, years-since-displacement indicators, and lost-job occupation and industry fixed effects; standard errors cluster by displacement year.

State regressions add state and displacement-year fixed effects; standard errors cluster by state.

State unemployment is available locally through 2019, so state columns are restricted to covered displacement years.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

observe an employed-worker job-finding probability, so I use the topical labor-market module's expected time to find an equivalent job and merge it forward to later quarterly labor-market observations for the same respondent. Restricting this merge to the nearest subsequent quarterly round gives the same qualitative conclusion.

B.7 2016 Election Event Study

This subsection provides an illustration of one source of within-location belief dispersion: partisan priors interacting with salient political events. Following Mian et al. (2018) and Coibion et al. (2020), the 2016 U.S. presidential election generated a sharp divergence in macroeconomic expectations across partisan-leaning groups. I use this episode to show that workers in similar local labor markets can hold systematically different macro views, and that these views can shift discontinuously around a major news event. I emphasize that this exercise is *illustrative*: it is not used as a stand-alone identification strategy for the causal effect of beliefs on search.

Design and sample. I classify states by whether the Republican or Democratic candidate won the state in November 2016. To focus on states with clearer partisan alignment, I drop swing states (defined based on the election margin threshold used in the replication code) and restrict attention to the 2016–2017 window. Because treatment is defined at the state level and the effective number of clusters is limited, results should be interpreted cautiously.

Table A17. External Validation in the ECB Consumer Expectations Survey

Panel A. Macro pessimism predicts perceived job-loss risk			
Outcome	Coef. on expected unemployment	SE	N
P(lose job in next 3 months)	0.1450^{***}	(0.0104)	212,582
P(lose job in next 3 months), with expectation controls	0.0887^{***}	(0.0185)	212,581
Panel B. Perceived job-loss risk predicts search			
Outcome	Coef. on P(lose job)	SE	N
Actively looking for a job	0.4736^{***}	(0.0089)	208,018
P(look for job in next 3 months)	0.6161^{***}	(0.0077)	176,293
Job-search intensity	0.3093^{***}	(0.0313)	25,711
Panel C. Supplementary interaction with expected time to find an equivalent job			
Outcome	Job-loss slope, stronger-finding group	Interaction with weak finding	N
Actively looking for a job	0.4367^{***}	-0.0306	31,151
	(0.0308)	(0.0504)	
P(look for job in next 3 months)	0.6229^{***}	-0.0289	26,730
	(0.0272)	(0.0438)	
Job-search intensity	0.4288^{***}	-0.5662^{***}	3,501
	(0.1183)	(0.1987)	

Notes: ECB CES probability variables are scaled to $[0, 1]$. The sample uses the ECB release dated 28 April 2026 and includes quarterly labor-market observations through 2026Q1 matched to monthly expectations. In Panel A, macro pessimism is the expected unemployment rate over the next 12 months. The second row adds controls for expected inflation, expected economic growth, current perceived unemployment, and expected household financial conditions. In Panel B, job-loss risk is the perceived probability of losing the current job in the next 3 months; these regressions also control for expected unemployment. Panels A and B use employed respondents in the regular quarterly labor-market module. Panel C uses the topical labor-market module's expected time to find an equivalent job, measured in round 41, merged forward to later regular quarterly rounds for the same respondent; it includes the same expectation controls as Panel B. "Weak finding" equals one if expected time to find an equivalent job is above the weighted median, i.e. 7 months or longer. All specifications use ECB CES weights, survey-round fixed effects, country fixed effects, demographic controls, and respondent-clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

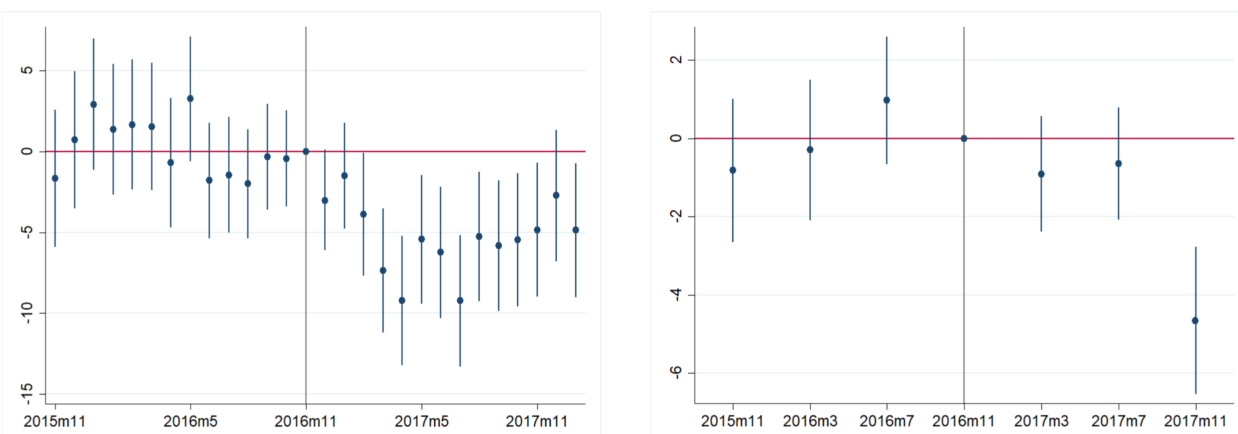
Event-study specification. I estimate an event-study that compares Republican-won and Democratic-won states around the election:

$$y_{ist} = \sum_{k \neq -1} \delta_k \left[\mathbb{1}\{t = k\} \times \text{RepWon}_s \right] + \mu_s + \lambda_t + \mathbf{X}'_{it}\gamma + \varepsilon_{ist}, \tag{13}$$

where t indexes event time (months relative to November 2016), μ_s and λ_t are state and month fixed effects, and $\mathbf{X}_{it}$ includes the baseline demographic controls. Standard errors are clustered at the state level.

Findings. Appendix Figure A3 shows two patterns. First, macro unemployment expectations diverge sharply after the election: respondents in Republican-won states report a persistent decline in macro pessimism relative to respondents in Democratic-won states, with little evidence of differential pre-trends. This illustrates that belief dispersion about aggregate conditions can be substantial even among workers facing similar local environments.

Second, the corresponding series for search hours is consistent with lower post-election search in Democrat-won states relative to Republican-won states, but is imprecisely estimated. Precision is limited because search hours are observed only in the LMM (every four months) and because the effective sample size is smaller once restricting to non-swing states. I therefore treat the search evidence as suggestive.



Macro Unemployment Expectations

Search Hours

Notes: The figure plots event-study coefficients comparing Republican- and Democrat-won non-swing states around the November 2016 election. The left panel shows divergence in macro unemployment expectations; the right panel shows corresponding movements in search hours. Outcomes are weighted; specifications include state and month fixed effects. The quarterly LMM limits precision, so this evidence is illustrative rather than definitive.

Figure A3. Event Study: 2016 Presidential Election

The difference-in-differences specification is

$$y_{ist} = \delta\,(\text{RepWon}_s \times \text{Post}_t) + \mathbf{X}_{it}'\gamma + \mu_s + \lambda_t + \varepsilon_{ist},$$

where $\text{Post}_t = \mathbb{1}\{t \geq \text{Nov 2016}\}$ and the sample is restricted to non-swing states over 2016–2017. Standard errors are clustered at the state level.

Table A18. 2016 Election (Appendix): DID (With Controls)

	(1) E(U exp)	(2) P(Lose)	(3) P(Leave)	(4) Search (arc)	(5) Search + beliefs
Rep-won × Post	-0.060***	0.014	0.034**	0.142	0.162
	(0.014)	(0.010)	(0.016)	(0.193)	(0.197)
Unemp exp					0.054
					(0.172)
P9: P(Lose)					0.856***
					(0.228)
N	20932	12256	12254	587	586

Notes: Difference-in-differences using Rep-won × Post (Nov 2016) for non-swing states, 2016–2017. Two-way fixed effects (state, month) with controls; standard errors clustered by state. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The difference-in-differences specification confirms a sharp post-election divergence in macro unemployment expectations across Republican- and Democratic-won states. For search, the positive Rep-won×Post estimate implies lower post-election search in Democrat-won states relative to Republican-won states, but the estimate is imprecise, reflecting the low frequency of the LMM and the limited number of state-level clusters. I interpret this exercise as an illustration of belief dispersion rather than a definitive estimate of the causal effect of macro beliefs on search.

Overall interpretation of robustness evidence. Across specifications, three patterns are stable:

1. The perceived job-loss probability consistently predicts higher search intensity.
2. The effect survives demanding geographic and sectoral fixed effects.
3. The relationship is present cross-sectionally and is broadly consistent with the within-person evidence, though the latter is imprecisely estimated due to limited LMM repeats.

These exercises do not identify causal belief shocks. They show that the empirical relationship linking perceived job-loss risk to search behavior is not easily explained by local shocks, industry conditions, functional-form choices, or fixed individual heterogeneity.

Together with the validation evidence in Section 2.4 (beliefs predict subsequent separations) and the mechanism evidence in Section 2.5 (stated motives and interaction patterns), the robustness exercises support the interpretation that employed workers adjust search effort in response to perceived separation risk.

C Model Appendix

This appendix collects calibration details, the full parameter list, derivations, and robustness exercises for the quantitative model.

C.1 Calibration Details

Table A19. Externally Set Parameters

Parameter	Description	Value	Source
β	Monthly discount factor	0.9967	4% annual rate
b	Flow value of unemployment	0.40	UI replacement benchmark
α	Matching elasticity with respect to unemployment	0.50	Petrongolo and Pissarides (2001)
ϕ	Worker bargaining share	0.50	Hosios benchmark
η	Employed/unemployed contact ratio	0.10	Employed search has lower time intensity
$\delta(B)/\delta(G)$	Recession-expansion job-loss probability ratio	1.67	Layoffs and discharges; Fujita and Ramey (2009)
$y(G), y(B)$	Productivity	1.00, 0.95	5% output gap
Π_{GG}, Π_{BB}	State persistence	0.99, 0.90	NBER durations
$m(G), m(B)$	Belief multipliers	1.00, 1.00	No state-dependent belief wedge

Notes: Parameters are monthly. The job-loss probability ratio governs cyclicality; the level of job-loss risk is internally calibrated after accounting for the fraction of offers usable as insurance.

Simulated-regression check. The main calibration targets the analytical two-state analogue of the empirical search semi-elasticity. I also solve the model, simulate worker-level

Table A20. Untargeted Fit Diagnostics

Moment	Data	Model	Interpretation
Employed search cyclicality, $s(B)/s(G)-1$	+4.4%: 3.71 to 3.87 hours	+6.4%: 2.990 to 3.182	Search rises in high-slack months and in the bad state.
Unemployment recession gap, $u(B)-u(G)$	+2.52 pp: 5.75% to 8.27%	+3.71 pp: 5.39% to 9.10%	The model recession is somewhat larger but in the empirical range.
U-to-E cyclicality, $p_{UE}(B)-p_{UE}(G)$	−0.87 pp: 20.5% to 19.6%	−1.48 pp: 28.2% to 26.7%	Job finding falls in high-slack months.
E-to-U inflow cyclicality, $p_{EU}(B)/p_{EU}(G)$	1.12: 1.10% to 1.23%	1.66: 1.61% to 2.67%	The model has stronger inflow cyclicality than raw SCE transitions.
Total E-to-E cyclicality, $p_{EE}(B)-p_{EE}(G)$	−4.34 pp: 5.95% to 1.61%	+0.08 pp: 3.83% to 3.91%	The model misses the procyclicality of total E-to-E mobility in this lower-bound SCE measure.

Notes: These moments are not used to estimate the SCE calibration. The data column splits SCE worker-months into high- and low-slack months, with high slack defined as months above the median monthly SCE unemployment share. The search row uses employed respondents with observed search hours. The transition rows use next-wave SCE transitions; the E-to-E row is the all-employed lower-bound probability, coding missing transition-supplement responses as no E-to-E transition. The model column compares the recession and expansion states in the SCE calibration. The model's average U-to-E probability is calibrated to the stock-flow object rather than to the raw next-wave SCE transition probability, so the U-to-E row should be read as a cyclical comparison rather than a level target.

observations, and run the same no-controls and controlled search regressions on model-generated data. This simulated-regression exercise is useful, but I do not use it as the headline target. In the current SCE calibration, the controlled simulated-regression coefficient is 2.60, above the empirical controlled coefficient of 1.69. A grid over γ can match the simulated-regression coefficient, but doing so implies a less elastic structural search response than the analytical target. I treat the simulated-regression exercise as a check on interpretation. It shows that the mapping from model effort to survey hours matters, while the qualitative conclusion remains the same: beliefs generate a meaningful search response at the worker level and a small effect on aggregate unemployment.

The additional moments in Table A20 show where the model fits the data and where it does not. In the SCE, employed search rises from 3.71 to 3.87 hours in high-slack months, a 4.4 percent increase. The model produces the same direction and a slightly larger response, 6.4 percent. Unemployment and job finding also move in the right direction. The main

Table A21. Alternative Calibrations

Profile	ρ_I	ρ_J	ω_J	$s(G)$	$s(B)$	E-to-E share
SCE core	0.532	0.108	0.145	2.990	3.182	0.094
Higher insurance use	0.750	0.102	0.145	2.043	2.235	0.146
CPS	0.223	0.079	0.053	1.959	2.152	0.050

Notes: The higher-insurance-use profile fixes $\rho_I = 0.75$ and recalibrates the remaining parameters. The CPS profile uses lower E-to-E and wage-gain targets. All profiles match the perceived-risk and search-response anchors.

miss is total E-to-E mobility: E-to-E mobility is procyclical in the SCE, while insured-separation moves make total E-to-E mobility slightly higher in the bad state of the model. I interpret the size of the SCE E-to-E decline cautiously because it is an all-employed lower-bound construction that codes missing transition-supplement responses as no E-to-E transition and is sensitive to the high-slack/COVID composition. The sign is consistent with the broader job-ladder literature, but the quantitative target should come from cleaner matched-employer or CPS-based evidence.

Appendix Table A22 lists the parameter values used in the quantitative exercises. Parameters are monthly unless otherwise noted.

Table A22. Full Parameter List

Parameter	Description	Value
β	Discount factor	0.9967
ϕ	Worker bargaining weight	0.5
α	Matching elasticity (unemployment)	0.5
b	Flow value of unemployment	0.4
$y(G), y(B)$	Productivity levels	1.00, 0.95
Π_{GG}, Π_{BB}	Markov persistence (monthly)	0.99, 0.90
$\delta(G), \delta(B)$	Job-loss probabilities (monthly)	0.024, 0.039
η	Employed/unemployed contact probability ratio	0.1
χ	Matching efficiency	0.338
κ	Vacancy posting cost	0.508
ψ	Search effectiveness	26.282
γ	Search cost curvature	0.0229
ζ	Belief level shifter	0.775

Notes: Parameters are monthly unless stated otherwise. The transition matrix has off-diagonal elements $\Pi_{GB} = 1 - \Pi_{GG}$ and $\Pi_{BG} = 1 - \Pi_{BB}$. Productivity is normalized to $y(G) \approx 1$. The baseline sets $m(G) = m(B) = 1$, so the cyclicality of perceived risk is inherited from objective separation risk rather than belief distortions.

C.2 Some Model Derivations

This appendix collects short derivations that clarify (i) the model's within-month timing, (ii) the search first-order condition when the worker evaluates the search margin using *perceived* job-loss probabilities, and (iii) the Nash wage condition used in the quantitative implementation.

Throughout, time is monthly and the aggregate state follows $z \in \{G, B\}$ with transition matrix $\Pi(z, z')$.

C.3 Monthly probabilities and within-month timing

Let $\delta(z)$ denote the *objective* monthly probability that the current job ends in state z:

$$\delta(z) \in (0,1), \qquad \text{with survival probability } 1 - \delta(z). \tag{14}$$

Employed search effort s maps into an offer probability

$$p_{\text{offer}}(z,s) = 1 - \exp\{-\eta\chi\theta(z)^{1-\alpha}\psi g(s)\}. \tag{15}$$

Ordering within the month. The implementation follows the timing in Section 3.1: the current aggregate state z is realized, workers choose search, offers and separations are realized within the month in that same state, and then the economy transitions to z' for continuation values. With this ordering, the *objective* probability of ending the month unemployed, given current state z and effort s, is

$$\delta_{EU}(z,s) = \left(1 - p_{\text{offer}}(z,s)\right)\delta(z). \tag{16}$$

Equivalently, offers reduce unemployment inflows by reducing the probability that a separation "hits" a worker who lacks a fallback option.

C.4 Perceived separation risk

Workers may perceive job-loss risk differently from the objective probability. In the implementation, the scaling applies to the monthly separation hazard. Let $\sigma(z)$ denote the hazard, with $\delta(z) = 1 - \exp\{-\sigma(z)\}$. Perceived risk is

$$\tilde{\sigma}(z) = \zeta \cdot m(z) \cdot \sigma(z), \qquad \tilde{\delta}(z) = 1 - \exp\{-\tilde{\sigma}(z)\}, \tag{17}$$

where $\zeta > 0$ is a level shifter and $m(z)$ allows perceived risk to vary differently across aggregate states. At monthly magnitudes, $\tilde{\delta}(z)$ is numerically close to scaling the probability itself, $\zeta m(z)\delta(z)$; the text uses whichever form reads more easily.

Under the same within-month ordering (search and offer realization within month, with separation risk realized at month-end), the worker's *perceived* probability of entering unemployment in current state z is

$$\tilde{\delta}_{EU}(z,s) = \big(1 - p_{\text{offer}}(z,s)\big)\,\tilde{\delta}(z). \tag{18}$$

This is the object that enters the worker's Bellman value and private search margin. Realized flows and firm survival are still governed by objective separation risk.

C.5 Deriving the search first-order condition

Fix current aggregate state z and let the worker choose effort $s(z) \equiv s$ after observing z. Let $W(z)$ and $U(z)$ denote the continuation values taken as given by the individual worker when choosing search. The private search margin evaluates the chance of entering unemployment using $\tilde{\delta}_{EU}$.

The local search objective can be written as

$$W(z) = w(z) - c(s) + \beta \sum_{z'} \Pi(z,z')\Big[\big(1 - \tilde{\delta}_{EU}(z,s)\big)W(z') + \tilde{\delta}_{EU}(z,s)\,U(z')\Big]. \tag{19}$$

Differentiate (19) with respect to s (holding $w(\cdot)$, $\theta(\cdot)$, and the continuation values $W(\cdot), U(\cdot)$ fixed from the perspective of the individual worker's static choice):

$$\begin{aligned} 0 &= -c'(s) + \beta \sum_{z'} \Pi(z,z')\Big[-\frac{\partial \tilde{\delta}_{EU}(z,s)}{\partial s} W(z') + \frac{\partial \tilde{\delta}_{EU}(z,s)}{\partial s} U(z')\Big] \\ &= -c'(s) + \beta\Big[-\frac{\partial \tilde{\delta}_{EU}(z,s)}{\partial s}\Big] \sum_{z'} \Pi(z,z')\big(W(z') - U(z')\big). \end{aligned} \tag{20}$$

Using (18), and noting that $\tilde{\delta}(z)$ does not depend on s,

$$\frac{\partial \tilde{\delta}_{EU}(z,s)}{\partial s} = -\tilde{\delta}(z)\,\frac{\partial p_{\text{offer}}(z,s)}{\partial s}. \tag{21}$$

Substitute (21) into (20) to obtain the search first-order condition:

$$c'(s(z)) = \beta\,\tilde{\delta}(z)\,\frac{\partial p_{\text{offer}}(z,s(z))}{\partial s}\,\sum_{z'} \Pi(z,z')\big(W(z') - U(z')\big). \tag{22}$$

Finally, with $p_{\text{offer}}(z,s) = 1 - \exp\{-\xi(z,s)\}$ and $\xi(z,s) = \lambda_e(\theta(z))\psi g(s)$,

$$\frac{\partial p_{\text{offer}}(z,s)}{\partial s} = \exp\big(-\xi(z,s)\big)\,\lambda_e(\theta(z))\psi g'(s) \;\geq 0. \tag{23}$$

Economic interpretation. Equation (22) shows the search margin. Search is attractive when (i) perceived job-loss risk $\tilde{\delta}(z)$ is high, (ii) effort moves offer availability (high $\partial p_{\text{offer}}/\partial s$), and (iii) the expected continuation surplus, $\sum_{z'} \Pi(z,z')\big(W(z') - U(z')\big)$, is large.

C.6 Nash wage with free entry

This subsection states the wage-setting condition used in the quantitative implementation and records the free-entry condition. Wages do not renegotiate within the month in response to an individual worker's search choice or outside offer.

Firm value of a filled job. Let $J(z)$ denote the firm's value of a filled job in state z, under the objective separation process. Because separations are realized within the current month in current state z, and because acceptable ladder offers create voluntary quits, the firm Bellman equation is

$$J(z) = y(z) - w(z) + \beta\left(1 - \delta(z)\right)\left(1 - \rho_J p_{\text{offer}}(z, s(z))\right) \sum_{z'} \Pi(z,z')\, J(z'). \tag{24}$$

Free entry. Let $V(z)$ be the value of a vacancy posted in state z. Free entry implies $V(z) = 0$. The monthly filling probability is $p_f(z) = 1 - \exp\{-\chi\theta(z)^{-\alpha}\}$. Posting a vacancy costs κ for the month; if the vacancy is filled, the firm obtains the continuation value of a filled job next month. The zero-profit condition is therefore

$$\kappa = \beta\, p_f(z) \sum_{z'} \Pi(z,z')\, J(z'), \qquad \text{with } p_f(z) = 1 - \exp\big\{-\chi\theta(z)^{-\alpha}\big\}. \tag{25}$$

Nash bargaining. Wages are set by Nash bargaining over the equilibrium match values, as in the main text. Let $X(z) = W(z) - U(z)$ denote the worker surplus and let $J(z)$ denote the firm value. Within the month, a change in the wage is a one-for-one transfer between the two parties: holding continuation values and the search policy fixed, $\partial X(z)/\partial w(z) = 1$ and $\partial J(z)/\partial w(z) = -1$. The wage solves

$$\max_{w(z)} \; X(z)^{\phi}\, J(z)^{1-\phi},$$

with first-order condition

$$(1 - \phi)\, X(z) = \phi\, J(z). \tag{26}$$

This condition is solved jointly with the worker values, the firm value, free entry, and the search first-order condition. In the textbook model without on-the-job search, combining (26) with free entry delivers the familiar closed form $w(z) = (1 - \phi)b + \phi\big(y(z) + \kappa\theta(z)\big)$. That closed form is not imposed in the headline job-ladder calibration, where wages are equilibrium unknowns. Because bargaining takes the search policy as given, the wage is not set strategically to discourage quitting; quit risk affects the bargain only through the level of $J(z)$.

C.7 Insurance-off algebraic comparison

The insurance channel is the difference between the unemployment inflow probability with and without offers being usable at separation time. Using (16) and the insurance-off counterpart $\delta_{EU}^{\text{off}}(z) = \delta(z)$, we have

$$\delta_{EU}^{\text{off}}(z) - \delta_{EU}(z, s) = \delta(z)\, p_{\text{offer}}(z, s). \tag{27}$$

Thus, the insurance value of search is larger when separations are more likely (high δ) and when offers are more likely to be in hand (high p_{offer}). In the job-ladder extension, the same comparison gives $\delta_{EU}^{\text{off}}(z) - \delta_{EU}(z, s) = \delta(z)\, \rho_I\, p_{\text{offer}}(z, s)$: shutting down the insurance channel removes only the usable share of offers.

C.8 Proofs for Section 3.6

This appendix gives the short arguments behind the analytical implications in Section 3.6. The logic comes from two objects: the search first-order condition (6) and the mapping from offers to unemployment transitions (7).

Proof of Proposition 1. Fix z and hold tightness, continuation values, and ladder values fixed. Let

$$A(z) \equiv \sum_{z'} \Pi(z, z')\rho_J \Delta_J(z'), \qquad B(z) \equiv \sum_{z'} \Pi(z, z')\rho_I\big(W(z') - U(z')\big).$$

Define

$$F\big(s, \tilde{\delta}(z)\big) \equiv c'(s) - \beta\, p_{\text{offer},s}(z, s)\big[(1 - \tilde{\delta}(z))A(z) + \tilde{\delta}(z)B(z)\big].$$

An interior optimum satisfies $F(s(z), \tilde{\delta}(z)) = 0$. By (4), $p_{\text{offer},s}(z,s) > 0$, so

$$F_{\tilde{\delta}} = -\beta\, p_{\text{offer},s}(z,s)\big[B(z) - A(z)\big] < 0.$$

The inequality uses the condition in the proposition, $B(z) > A(z)$. Also,

$$p_{\text{offer},ss}(z,s) = \exp\Big(-\lambda_e(\theta(z))\psi g(s)\Big)\,\lambda_e(\theta(z))\psi\Big[g''(s) - \lambda_e(\theta(z))\psi\,(g'(s))^2\Big] < 0,$$

because $g'(s) > 0$, $g''(s) \le 0$, and $\lambda_e(\theta(z))\psi > 0$. Hence

$$F_s = c''(s) - \beta\, p_{\text{offer},ss}(z,s)\big[(1-\tilde{\delta}(z))A(z) + \tilde{\delta}(z)B(z)\big] > 0.$$

By the implicit function theorem,

$$\frac{\partial s(z)}{\partial \tilde{\delta}(z)} = -\frac{F_{\tilde{\delta}}}{F_s} > 0.$$

□

Proof of Proposition 2. From (7), $\delta_{EU}(z,s) = \delta(z)(1 - \rho_I p_{\text{offer}}(z,s))$. Because $p_{\text{offer}}(z,s)$ is strictly increasing in s, $\delta_{EU}(z,s)$ is strictly decreasing in s. For a given offer probability, $\delta_{EU}(z,s)$ is increasing in the objective job-loss probability $\delta(z)$. □

Proof of Proposition 3. Let

$$\mathcal{MB}(z;s) = p_{\text{offer},s}(z,s)\sum_{z'}\Pi(z,z')\left[(1-\tilde{\delta}(z))\rho_J\Delta_J(z') + \tilde{\delta}(z)\rho_I\big(W(z') - U(z')\big)\right].$$

Define $F_z(s) \equiv c'(s) - \beta\,\mathcal{MB}(z;s)$. Under Assumption 1, $c'(s)$ is strictly increasing. Since $p_{\text{offer},ss}(z,s) < 0$ under the maintained search technology,

$$\frac{\partial \mathcal{MB}(z;s)}{\partial s} = p_{\text{offer},ss}(z,s)\sum_{z'}\Pi(z,z')\left[(1-\tilde{\delta}(z))\rho_J\Delta_J(z') + \tilde{\delta}(z)\rho_I\big(W(z') - U(z')\big)\right] < 0.$$

Therefore,

$$F_z'(s) = c''(s) - \beta\,\frac{\partial \mathcal{MB}(z;s)}{\partial s} > 0,$$

so $F_z(s)$ is strictly increasing and has a unique root.

If $\mathcal{MB}(B;s) \ge \mathcal{MB}(G;s)$ in a neighborhood containing the relevant effort choices, then $F_B(s) \le F_G(s)$ in that neighborhood. Since $F_G(s(G)) = 0$, it follows that $F_B(s(G)) \le 0$. Because F_B is increasing and crosses zero at $s(B)$, we must have $s(B) \ge s(G)$. □

Proof of Proposition 4. If $s^{\text{base}}(B) > s^{\text{fix}}(B)$, then by Proposition 2, $\delta_{EU}(B, s^{\text{base}}(B)) < \delta_{EU}(B, s^{\text{fix}}(B))$. With the aggregate state held fixed at B, flow balance between unemployment inflows and outflows implies the steady-state rate

$$u^{ss}(B) = \frac{\delta_{EU}(B, s)}{\delta_{EU}(B, s) + p_u(B)}, \tag{28}$$

which is strictly increasing in $\delta_{EU}(B, \cdot)$ for fixed $p_u(B)$, since $\partial u^{ss}/\partial \delta_{EU} = p_u/(\delta_{EU} + p_u)^2 > 0$. Steady-state unemployment is therefore lower under baseline search than under fixed search. □

C.9 Counterfactual Details

Each reported number re-solves the state-contingent equilibrium under the relevant calibration and then computes the fixed-search recession comparison using recession-state contact probabilities.

Table A23. Counterfactual Details

Calibration	Exercise	Dampening
SCE core	Job-ladder calibration	0.046 pp
Higher insurance use, $\rho_I = 0.75$	More offers usable at separation	0.115 pp
CPS	CPS calibration	0.030 pp

Notes: Dampening is $100 \times [u_B^{\text{fixed search}} - u_B^{\text{reoptimized}}]$. The higher-insurance-use case fixes $\rho_I = 0.75$ because self-reported job-loss E-to-E transitions plausibly understate the share of outside options that can protect workers at separation.

Table A24. Shutting Off the Insurance Use of Offers

Calibration	$s(B)$	$s(B)$ off	$u(B)$	$u(B)$ off	$\Delta u(B)$	$\Delta p_{EU}(B)$
SCE core	3.182	2.170	9.10%	10.77%	1.67 pp	0.55 pp
Higher insurance use	2.235	1.326	9.08%	11.68%	2.59 pp	0.87 pp
CPS	2.152	1.259	9.07%	9.67%	0.60 pp	0.21 pp

Notes: The insurance-off equilibrium is re-solved after setting outside offers to be unusable at separation. Ordinary ladder moves remain active. The table reports recession-state objects; changes are relative to the corresponding calibrated row.

C.10 Unemployment Insurance Counterfactual

This exercise raises unemployment benefits by 10 percent in the recession state and re-solves the equilibrium. Higher UI affects unemployment through two forces. The standard force is that higher benefits compress match surplus, raise wages, and lower tightness. The

Table A25. Advance-Information Upper-Bound Counterfactual

Calibration	$s(B)$	$s(B)$ notice	$\Delta s(B)$	$u(B)$	$u(B)$ notice	$\Delta u(B)$
SCE core	3.182	3.333	4.73%	9.10%	8.18%	-0.93 pp
Higher insurance use	2.235	2.356	5.42%	9.08%	7.62%	-1.46 pp
CPS	2.152	2.264	5.21%	9.07%	8.72%	-0.35 pp

Notes: The exercise re-solves the GE model after setting the advance-notice share to 0.475 and the notice window to 2.77 months, based on SCE job-loss/anticipated-layoff transitions. Advance notice affects only the offer-insurance protection technology: workers with notice have more time to receive an offer before separation. This is an upper-bound counterfactual, not part of the baseline calibration.

channel emphasized in this paper is narrower: higher benefits also reduce the insurance value of securing an offer before job loss, so they crowd out precautionary employed search. To isolate this search channel, I also re-solve the UI economy while holding search fixed at its baseline state-contingent level.

Table A26. Recession UI Increase and Offer-Insurance Crowd-Out

Profile	ρ_I	$\Delta s(B)$	Δ covered E-E	Search part	$\Delta u(B)$	Search part
Job-ladder extension, SCE core	0.532	-2.52%	-0.020 pp	-0.006 pp	0.284 pp	0.015 pp
Higher insurance use	0.750	-2.58%	-0.032 pp	-0.010 pp	0.324 pp	0.029 pp
CPS	0.224	-2.55%	-0.008 pp	-0.003 pp	0.243 pp	0.005 pp

Notes: The experiment raises the unemployment benefit flow by 10 percent in the recession state only and re-solves the GE model. Covered E-E is the monthly job-loss-related E-E probability in the recession state. The two "Search part" columns subtract the fixed-search UI equilibrium from the endogenous-search UI equilibrium, where fixed search holds both state-contingent search levels at their baseline values. Thus the unemployment search part isolates the extra unemployment generated by UI-induced search crowd-out, net of the standard UI surplus/tightness channel.

Table A26 shows the policy tradeoff. The search response is similar across rows because UI lowers the insurance value of search in each calibration. The composition effect differs sharply. Total covered E-to-E falls four times as much in the high-insurance row as in the CPS row, and the isolated search component shows the same ordering. The unemployment effect, however, is mostly not this paper's channel: of the 0.24–0.32 percentage point increase in recession unemployment, only 0.005–0.029 percentage points comes from search crowd-out. The rest is the standard UI effect operating through surplus, wages, and tightness. UI crowd-out of insured moves is economically ordered by ρ_I, but its contribution to recession unemployment is a small slice of the total UI effect.

Table A27. Idiosyncratic Belief Dispersion and the Search-Risk Gradient

Simulation support	Model slope	Empirical slope	Ratio	SD of perceived risk
Full SCE support	2.784	1.688	1.65	0.232
Central 80 percent	2.376	1.688	1.41	0.163
Interquartile range	1.276	1.688	0.76	0.083

Notes: This diagnostic adds idiosyncratic perceived-risk dispersion as a simulation layer while holding aggregate prices fixed. Workers draw perceived annual job-loss risk from the SCE cross-section and receive the model search policy implied by that belief. The empirical slope is the controlled SCE regression coefficient of weekly search hours on perceived job-loss probability. The local calibrated slope equals the empirical target, but the full-support simulated cross-section is steeper because the search policy is nonlinear over the wide SCE belief distribution.

C.11 AR(1) State-of-the-World Extension

As a check on the two-state aggregation, I also solve an extension in which the aggregate state is a discretized AR(1) state-of-the-world process,

$$\log y_t = \rho \log y_{t-1} + \varepsilon_t.$$

I use a seven-state Rouwenhorst approximation. The search and offer-arrival block is tied to the SCE/FMST evidence used in the main calibration (Faberman et al., 2022). The AR(1) process is a persistence check rather than a separately estimated productivity process. Job-loss risk is mapped into the state of the world so that a five percent decline in the state implies the same job-loss probability ratio as in the two-state calibration.

I recalibrate the AR(1) grid to the main average moments. The recalibration hits offer receipt and average E-to-E mobility closely, and comes close on unemployment and U-to-E flows. It does not fully recover the targeted search-belief semi-elasticity: the local AR(1) value is 1.38, compared with the target 1.69. I therefore read Table A28 as a disciplined robustness exercise rather than a replacement for the main two-state calibration.

C.12 Ladder Value Sensitivity

A natural concern is that the ladder value is understated. The headline calibration maps the observed wage gain from E-to-E moves into a one-period value gain. If these gains are persistent, this mapping may make ladder search too weak. Table A29 varies ω_J by factors of two and five, holding the SCE core ρ_I and ρ_J fixed and re-solving the state-contingent equilibria.

Increasing the ladder value raises search in both states, especially in expansions, and reduces the aggregate stabilization attributed to precautionary re-optimization. This goes

Table A28. AR(1) State-of-the-World Extension

Object	Good state	Bad state	Difference
Productivity/state y	1.000	0.949	-0.051
Search effort s	0.477	0.510	0.033
Offer probability	0.330	0.322	-0.009
U-to-E probability	0.282	0.262	-0.020
Unemployment	5.30%	9.23%	3.93 pp
Unemployment, fixed search	–	9.33%	–
Dampening, PE	–	0.098 pp	2.44%

Notes: The good state is the grid point with $y = 1$. The bad state is the grid point closest to a five percent decline. Fixed search holds bad-state employed search at the good-state level while keeping bad-state contact probabilities fixed, so the dampening entry is a partial-equilibrium comparison.

Table A29. Sensitivity to the Ladder Value

ω_J multiplier	ω_J	$s(G)$	$s(B)$	$p_{\text{offer}}(G)$	$p_{\text{offer}}(B)$	Dampening
1	0.145	2.990	3.182	0.330	0.322	0.046 pp (1.24%)
2	0.290	3.800	3.884	0.366	0.351	0.016 pp (0.44%)
5	0.724	5.570	5.495	0.428	0.406	-0.010 pp (-0.29%)

Notes: The sensitivity varies the ladder gain parameter holding the SCE core ρ_I and ρ_J fixed and re-solves the state-contingent equilibria. Larger ladder values make employed search less precautionary at the margin, so the fixed-search unemployment comparison shrinks.

against the concern that the small aggregate effect is caused by an undervalued ladder motive. A stronger ladder motive makes total search larger but makes the marginal cyclical insurance channel smaller.

Insurance-off benchmark. In the job-ladder model, shutting down the insurance use of offers corresponds to setting $\rho_I = 0$. Then the realized employment-to-unemployment probability becomes

$$p_{EU}^{\text{off}}(z) = \delta(z),$$

independent of employed search. Ladder search may still be valuable through ρ_J and ω_J, but it does not directly reduce unemployment inflows. This is why the aggregate unemployment margin in the main text is governed by ρ_I, while total E-to-E mobility is governed jointly by ρ_I and ρ_J.